Asymptotic Behavior of Recursive State Estimations With Intermittent Measurements

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***Abstract*—Convergence rate and stationary distribution approximation are investigated in this paper for both the covariance matrix of the Kalman filter and the pseudo-covariance matrix of a robust recursive state estimator developed by us. When the measurement dropping process is described by a Markov chain and an associated plant is both controllable and observable, it is proved that if the dropping probability is less than 1, these two matrices converge exponentially to a stationary distribution independent of its initial values. Moreover, when they are initialized with the stabilizing solution of an associated algebraic Riccati equation, both of them are shown to converge to an ergodic process. Based on these results, two approximations are derived for their stationary probability distributions, as well as a bound of approximation errors. It has been made clear that rather than the delta function suggested by Kar and Moura, a series of delta functions give a more accurate approximation, and replacement of these estimators' update gain by a constant matrix usually deteriorates their steady estimation accuracy. A numerical example is provided to illustrate the obtained theoretical results.**

***Index Terms*—Ergodicity, network system, random measurement dropping, recursive state estimation, stationary distribution.**

I. Introduction

WITH the development of network technologies, numerous novel anticipations, as well as various new technical issues, rise in system analysis and synthesis, due to the significant differences in information exchange methods between traditional and network systems. Transmission quality of a communication channel has significant influences on the performance of a network system, as wireless link or utilization of commercial channels introduces many possibly detrimental factors, such as random data delay and/or loss, etc. These factors may result in significant system performance degradations, and even destroy system stability in the worst situation. To clarify and reduce influences of these factors, reinvestigations are required for many system design methodologies. Among them, one important issue is state estimation with random measurement droppings, in which plant output measurements are stochastically lost due to communication failures [2], [5], [19], [23].

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Over the last decade, this problem has attracted extensive attention and various results have been obtained. On the basis of a separation principle, it is proved in [5] that for an optimal control system, the intermediate nodes in state estimations do not need to access the plant control inputs, which enables the development of a recursive optimal state estimator. Some relations are established in [17] between a constant gain Kalman filter and switched estimators. A recursive state estimation procedure is developed in [6], in which both the current and the previous plant output measurements are transmitted. In [19], optimality of the traditional Kalman filter is established under the existence of random measurement droppings, provided that information is available in the received data on whether or not it is a measured plant output. It has also been made clear that for an unstable plant, to guarantee boundedness of the expectation of the covariance matrix of estimation errors (CMEE), in addition to controllability and observability, the measurement dropping probability must be lower than some threshold values. Some necessary and sufficient conditions have been obtained for this boundedness of some particular classes of unstable systems, which depend on the spectral radius of the plant state transition matrix (STM). It is observed, however, that although complete loss of plant output measurements usually has an essential zero probability, it is the dominating fact that leads to an infinite expectation of this CMEE [16]. This observation motivates investigations on its probability distribution, which is argued to be a more appropriate measure on the performance of a state estimator with random data missing [12], [16].

Particularly, using a Stieltjies transformation, some explicit formulas are obtained in [21] that enable computations of the steady eigenvalue distribution of the aforementioned CMEE for a stable plant. Some upper and lower bounds are derived respectively in [15], [16] for the probability that this CMEE is smaller than a prescribed positive definite matrix (PDM). Under the condition that an unstable plant has a diagonalizable STM, [12] shows that if some controllability and observability conditions are satisfied, the trace of this CMEE decays according to a power law. Based on the contractive properties of Riccati recursions and convergence conditions on random iterated functions, this CMEE is proved in [2] to converge in general to a stationary distribution independent of its initial values. When the observation arrival is modeled by a Bernoulli process and the packet arrival probability is approximately equal to 1, the stationary distribution of this CMEE is shown in [9] to converge weakly to the delta function centered around the unique stabilizing solution of an associated deterministic Riccati equation, and the convergence rate satisfies a moderate deviation principle with a good rate function.

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When a plant model is not accurate, which is the general situation in solving an engineering problem, recursive state estimations that are robust against modelling errors have also been extensively investigated, for example, [4], [8], [14], [18], [22], and the references therein. Some of these methods have already been extended to systems with an imperfect communication channel, such as the works in [13], [23]. Particularly, a recursive robust state estimator is derived in [23] penalizing sensitivity of an innovation process to modelling errors, which reduces to the Kalman filter when the plant model is accurate. Moreover, some necessary and sufficient conditions have also been established on the convergence of its pseudo-covariance matrix (PCM) to a stationary distribution.

These investigations have made many important theoretical issues clear about state estimations with random measurement arrivals, and the obtained results appear greatly helpful in the analysis and synthesis of network systems. Some important issues of this estimation problem, however, still need further efforts. Among them, one essential problem is about the convergence rate of the CMEE in the Kalman filtering or the PCM in the robust estimations, and a more accurate characterization of their stationary distributions. Note that no explicit forms are given for this stationary distribution in [2], [23], and there are still no efficient methods in determining the good rate function of [9] which is required in obtaining the best parameter of the convergence rate. Similar to those of the conventional Kalman filter [8], [18], convergence rate of these random matrices describes influences of their initial values on estimation accuracies which are usually not accurate, and their stationary distribution directly connects to steady estimation accuracies and properties of the steady update gain matrices. Settlement of this problem is essential in determining various important characteristics of a networked system, such as quality requirements on the communication channel, possibilities of replacing the Kalman filter or the robust state estimator with a time invariant observer, etc. The former is important in network system design, while the latter is very attractive in reducing computational costs of a state estimator.

This paper reinvestigates properties of the stationary distribution of the CMEE of the Kalman filter and the PCM of the robust state estimator of [23]. The data missing process is assumed to be described by a Markov chain, which can include the Bernoulli process adopted in [9] as a special case. On the basis of the Riemannian metric defined in [1] for the space of PDMs and the central limit theorem established in [10] for Markov chains, it is proved that when an associated plant is both controllable and observable, these matrices converge to a stationary distribution, provided that the data arrival probability is greater than zero. Rather than a power law, the convergence rate is proved to be exponential. It has also been shown that when these matrices are initialized with the stablilizing solution of an associated algebraic Riccati equation, both of them are stationary and ergodic. From these results, two approximations are given for their stationary distributions, as well as their convergence rate to the actual value. It has been made clear that compared with the delta function suggested in [9], a series of delta functions are more appropriate in approximating these stationary distributions. Moreover, a simple replacement of the update gain of these state estimators with a constant matrix usually sacrifices steady estimation accuracy, which is significantly different from the conventional Kalman filter.

Compared with the results of [23], major contributions of this paper lie in its clear description of the convergence rate, and explicit formulas for two approximations of the stationary distribution that can be simply computed.

The outline of this paper is as follows. At first, in Section II, the problem formulation is provided, together with some preliminary results on Markov processes and Riccati recursions. Afterwards, stationarity and ergodicity properties of the CMEE/PCM process are investigated in Section III, while Section IV derives an approximation of the stationary distribution of this CMEE/PCM, as well as its convergence rate to the actual value. A numerical example is provided in Section V to illustrate the effectiveness and accuracy of the suggested approximation methods. Finally, some concluding remarks are given in Section VI. An Appendix is included to give proofs of some technical results.

The following notation and symbols are adopted. The product $\Phi_{k_1}\Phi_{k_1-1 \text{ or } k_1+1}\cdots\Phi_{k_2}$ is denoted by $\prod_{j=k_1}^{k_2}\Phi_j$, while the transpose of a matrix/vector is indicated by the superscript T. For matrices P and $\Phi = [\Phi_{ij}|_{i,j=1}^{2}]$ with compatible dimensions, a Homographic transformation $\mathbf{H}_m(\Phi, P)$ is defined as $\mathbf{H}_m(\Phi, P) = [\Phi_{11}P + \Phi_{12}][\Phi_{21}P + \Phi_{22}]^{-1}$. $\mathbf{P}_r(\cdot)$ is used to denote the probability of the occurrence of a random event, while $\mathbf{V}_{ar\{\dagger\}}\{*\}$ and $\mathbf{E}_{\{\dagger\}}\{*\}$ the variance of a random variable $*$ and the mathematical expectation of a random matrix valued function (MVF) $*$ with respect to the random variable $\dagger$. The subscript $\dagger$ is usually omitted when it is obvious. $O(x)$ stands for a number that is of the same order in magnitude as x, while $\phi(t)$ the distribution function of a normally distributed random variable with its mathematical expectation and variance respectively being 0 and 1. $I_{\mathcal{A}}(x)$ is the indictor function which equals 1 when x belongs to the set $\mathcal{A}$ and zero elsewhere, and $\#\{*\}$ the number of elements in a set, $\lceil * \rceil$ the minimal integer that is not smaller than $*$. To avoid complicated expressions, the same symbol is used for a random variable/process and its realization in this paper.

II. Problem Description and Some Preliminaries

Let α and β be two constants belonging to (0, 1), $A^{[0]}$ and $A^{[1]}$ two prescribed real square matrices with an equal dimension, and $G^{[0]}$, $G^{[1]}$ and $H^{[1]}$ other three known real matrices with compatible dimensions. Consider the following random matrix recursion

$$P_{k+1|k+1} = \begin{cases} A^{[0]}P_{k|k}A^{[0]T} + G^{[0]}G^{[0]T} & \gamma_{k+1} = 0 \\ \left\{\left(A^{[1]}P_{k|k}A^{[1]T} + G^{[1]}G^{[1]T}\right)^{-1} \right. & \\ \quad \left. + H^{[1]T}H^{[1]}\right\}^{-1} & \gamma_{k+1} = 1 \end{cases} \tag{1}$$

starting from an initial PDM $P_{0|0}$, in which γ_k is a Markov chain described by

$$\begin{bmatrix} \mathbf{P}_r(\gamma_k = 1) \\ \mathbf{P}_r(\gamma_k = 0) \end{bmatrix} = \begin{bmatrix} \alpha & 1-\beta \\ 1-\alpha & \beta \end{bmatrix} \begin{bmatrix} \mathbf{P}_r(\gamma_{k-1} = 1) \\ \mathbf{P}_r(\gamma_{k-1} = 0) \end{bmatrix} \tag{2}$$

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Here, $\gamma_k = 1$ means that the plant output measurement at the k-th time instant has been successfully transmitted, while $\gamma_k = 0$ represents a data transmission failure of the communication channel.

The objectives of this paper are to study convergence rate of this random matrix recursion to a stationary distribution, as well as approximations of this distribution.

When $A^{[0]} = A^{[1]}$ and $G^{[0]} = G^{[1]}$, the matrices $A^{[1]}$, $G^{[1]}$ and $H^{[1]}$ are respectively the plant STM, input matrix and output matrix. In this case, the above recursion becomes that of the CMEE of the Kalman filter for a linear time invariant plant with intermittent observations. That is, the matrix $P_{k|k}$ is in fact its CMEE. This recursion is originally derived in [19] and its properties are discussed extensively, for example, [2], [9], [12], [16]. When $A^{[0]} \neq A^{[1]}$ and/or $G^{[0]} \neq G^{[1]}$, the matrices $A^{[0]}$ and $G^{[0]}$ represent nominal values of the plant STM and input matrix, while $A^{[1]}$, $G^{[1]}$ and $H^{[1]}$ modifications of these matrices and the plant nominal output matrix, in which derivatives of plant parameters to modelling errors have been taken into account [22], [23]. Under such a situation, this recursion corresponds to that of the PCM of the robust state estimator developed in [23] when both the plant nominal model and the derivatives of the plant parameters to modelling errors are time invariant. In other words, the matrix $P_{k|k}$ stands now for the PCM of this robust estimator. In either of these two situations, the matrices $A^{[*]}$, $G^{[*]}$ with $* = 0$, 1 and $H^{[1]}$, are completely determined by plant nominal model parameters and therefore known. As the matrix $P_{k|k}$ directly connects to both the estimation accuracy and the update gain matrix of these two estimators, its asymptotic properties are very important in understanding their characteristics.

It is worthwhile to point out that when $A^{[0]} = A^{[1]}$ and $G^{[0]} = G^{[1]}$ are not satisfied, the approaches adopted in [2], [9], [12], [16] can not be utilized in analyzing the asymptotic properties of the above matrix recursion, as these two equalities are essential in deriving results of these papers. More specifically, [2, Equation (3)], [12, Proposition 1], [16, Equation (6)], [9, Equation (48)] are no longer valid without satisfaction of these two equalities, which are starting points of these papers.

In studying asymptotic properties of Riccati recursions, an efficient metric between two PDMs is the Riemannian distance defined in [1]. Compared to the trace or induced norm adopted in [9], [12], [16], [19], an attractive property of this metric is that both Riccati recursions and Lyapunov recursions are contractive. This metric has been used in establishing convergence respectively in [1] for the Kalman filter with random plant parameters, in [2] for the Kalman filter with random data dropping, and in [23] for a robust state estimator. More precisely, let P and Q be two $n \times n$ dimensional PDMs and λ_i an eigenvalue of the matrix PQ^{-1}. Then, the Riemannian distance between these two matrices, denote it by $\delta(P, Q)$, is defined as $\delta(P,Q) = \sqrt{\sum_{i=1}^{n} ln^2(\lambda_i)}$. In combination with properties of Hamiltonian matrices and the Homographic transformations $\mathbf{H}_m(\Phi, X)$, this metric plays an essential role in the following analysis on the asymptotic properties of the random process $P_{k|k}$.

To analyze asymptotic properties of this random process, the following results on Markov processes and the Homographic transformation $\mathbf{H}_m(\Phi, X)$ are also needed.

Lemma 1. [10], [20]: Let $x_i|_{i=0}^{\infty}$ be a positive recurrent irreducible Markov chain defined on a probability space $(\Omega, \mathcal{F}, P)$ with a countable state space $\mathcal{I}$, and $f(\cdot)$ be a real valued function defined on $\mathcal{I}$. Denote the time of the α-th entrance of the Markov chain into its j-th state by $\tau_\alpha^{[j]}$, and $\sum_{k=\tau_\alpha^{[j]}}^{\tau_{\alpha+1}^{[j]}-1} f(x_k)$ by $f_\alpha^{[j]}$. If both $\mathbf{E}(|f_\alpha^{[j]}|^3)$ and $\mathbf{E}(|\tau_{\alpha+1}^{[j]} - \tau_\alpha^{[j]}|^3)$ are finite, and $\sigma_j = \sqrt{\mathbf{V}_{ar}\{f_\alpha^{[j]} - s(f)(\tau_{\alpha+1}^{[j]} - \tau_\alpha^{[j]})\}}$ is greater than 0, then

$$\sup_{t\in\mathcal{R}} \left| \mathbf{P}_r \left\{ \frac{1}{\sigma_j \sqrt{n\pi_j}} \left(\sum_{k=0}^{n} f(x_k) - (n+1)s(f) \right) < t \right\} - \phi(t) \right| = O\left(\left(\frac{ln(n)}{n} \right)^{\frac{1}{4}} \right) \quad (3)$$

in which $s(f) = \sum_{i\in\mathcal{I}} (f(i)/\mu_i)$ with μ_i the mathematical expectation of the recurrence time of the i-th state, and $\pi_i = \mu_i^{-1}$.

Lemma 2. [3]: Assume that a Markov process $x_i|_{i=0}^{\infty}$ has an unique stationary distribution μ. Then, this process is ergodic with x_0 taking any element of the support of the distribution μ.

Lemma 3. [8], [23]: Assume that matrices Φ_1, Φ_2 and P have compatible dimensions. Moreover, assume that all the required matrix inverses exist. Then, $\mathbf{H}_m(\Phi_2, \mathbf{H}_m(\Phi_1, P)) = \mathbf{H}_m(\Phi_2\Phi_1, P)$.

III. Asymptotic Properties of the Random Matrix Recursion

To analyze asymptotic properties of the random process $P_{k|k}$, the following 3 conditions are adopted throughout this paper.

- Condition I: Both of the matrices $A^{[0]}$ and $A^{[1]}$ are invertible.
- Condition II: $(A^{[1]}, G^{[1]})$ is controllable and $(A^{[1]}, H^{[1]})$ is observable.
- Condition III: $0 < \alpha, \beta < 1$.

Note that the measure of singular matrices is equal to zero [7]. It can therefore be declared that Condition I is usually satisfied. In addition, even if it is not satisfied, a small perturbation on the elements of the matrices $A^{[0]}$ and/or $A^{[1]}$ can make it satisfied. As the results of this paper do not explicitly depend on either the inverse of $A^{[0]}$ or that of $A^{[1]}$, validity of the corresponding conclusions can be proved through taking their limit with reducing the perturbation magnitude to zero. On the other hand, recall that observability is necessary for plant state reconstruction from its input/output data, and controllability is necessary for positioning plant poles into a desirable area that is important in stabilizing a state estimator [8]. It appears safe to claim that Condition II is not very restrictive for guaranteeing the convergence of the random matrix recursion of (1). Moreover, when $\alpha = 1$ and $\beta = 0$, this recursion reduce to that of state estimation without data loss, while $\alpha = 0$ and $\beta = 1$ means that plant output measurement is never sent to the state estimator. The former has been well studied in traditional Kalman filtering and robust state estimations, while the latter has very limited engineering significance. If $\alpha = \beta = 1$, depending on the initial value of the Markov chain, the recursion of (1) becomes one of the aforementioned

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two scenarios. Furthermore, the results of this paper remain valid when $\alpha = \beta = 0$, which can be understood from their proofs. This situation is not included in this paper only to avoid complicated statements.

These discussions suggest that the above 3 conditions are not very restrictive from a system engineering point of view.

When Condition I is satisfied, define matrices $M^{[0]}$ and $M^{[1]}$ respectively as

$$M^{[0]} = \begin{bmatrix} A^{[0]} & G^{[0]}G^{[0]T}A^{[0]-T} \\ 0 & \left(A^{[0]}\right)^{-T} \end{bmatrix}$$

$$M^{[1]} = \begin{bmatrix} A^{[1]} & G^{[1]}G^{[1]T}\left(A^{[1]}\right)^{-T} \\ H^{[1]T}H^{[1]}A^{[1]} & \left[I + H^{[1]T}H^{[1]}G^{[1]}G^{[1]T}\right]\left(A^{[1]}\right)^{-T} \end{bmatrix}$$

Then, direct algebraic manipulations show that $P_{k+1|k+1}$ of (1) can be reexpressed as

$$P_{k+1|k+1} = \begin{cases} \mathbf{H}_m\left(M^{[0]},\ P_{k|k}\right) & \gamma_{k+1} = 0 \\ \mathbf{H}_m\left(M^{[1]},\ P_{k|k}\right) & \gamma_{k+1} = 1 \end{cases} \tag{4}$$

A matrix Φ is called Hamiltonian if it satisfies $\Phi^T J \Phi = J$, in which $J = [0\ I; -I\ 0]$ [1], [8], [23]. From the definitions of $M^{[0]}$ and $M^{[1]}$, direct algebraic manipulations show that both of them are Hamiltonian. It can therefore be declared from [1, Proposition 1.5] or [23, Lemma 2] that for any PDM X with a compatible dimension, both $\mathbf{H}_m(M^{[0]}, X)$ and $\mathbf{H}_m(M^{[1]}, X)$ are well defined and positive definite. Moreover, a repetitive utilization of Lemma 3 directly leads to that if $P_{0|0}$ is positive definite which is generally satisfied in practical engineering problems, then, for any positive integer j smaller than k

$$\begin{aligned} P_{k|k} &= \mathbf{H}_m\Big(M^{[\gamma_k]},\ \mathbf{H}_m\Big(M^{[\gamma_{k-1}]}, \cdots, \\ &\qquad \mathbf{H}_m\left(M^{[\gamma_{j+1}]}, P_{j|j}\right)\cdots\Big)\Big) \\ &= \mathbf{H}_m\left(\prod_{i=k}^{j+1} M^{[\gamma_i]},\ P_{j|j}\right) \end{aligned} \tag{5}$$

When $j = 0$, this equation reveals a relation between $P_{k|k}$ and its initial value $P_{0|0}$. Furthermore, define α_{1h} and α_{0h}, respectively, as

$$\alpha_{0h} = \sup_{X,Y>0,\ X\neq Y} \frac{\delta\left(\mathbf{H}_m(M^{[0]}, X),\ \mathbf{H}_m\left(M^{[0]}, Y\right)\right)}{\delta(X,Y)}$$

$$\alpha_{1h} = \sup_{X,Y>0,\ X\neq Y} \frac{\delta\left(\mathbf{H}_m\left(M^{[1]}, X\right),\ \mathbf{H}_m\left(M^{[1]}, Y\right)\right)}{\delta(X,Y)}$$

It has been proved in [1], [23] that $\alpha_{0h} \leq 1$ and $\alpha_{1h} < 1$ are valid whenever Conditions I and II are satisfied.[1]

Based on these properties and Lemma 1, the following theorem is obtained. Its proof is given in the Appendix.

Theorem 1: For a prescribed binary sequence $\gamma_i|_{i=1}^{k}$ with $\gamma_i \in \{0, 1\}$, define functions $\Phi_k(\cdot)$ and $\rho_k(\cdot,\cdot)$ on the set of PDMs respectively as $\Phi_k(X) = \mathbf{H}_m(M^{[\gamma_k]}, \mathbf{H}_m(M^{[\gamma_{k-1}]}, \cdots, \mathbf{H}_m(M^{[\gamma_1]}, X)\cdots))$ and $\rho_k(X,Y) = \delta(\Phi_k(X), \Phi_k(Y))$. Assume that Conditions I–III are simultaneously satisfied. Then, for the random process $\gamma_i|_{i=1}^{\infty}$ and arbitrary PDMs X and Y

$$\lim_{k\to\infty} \rho_k(X,Y) = 0, \quad \text{in probability} \tag{6}$$

Moreover, the convergence rate is $O(e^{-\sqrt{k}})$.

For a prescribed PDM X and a particular realization $\gamma_k|_{k=1}^{\infty}$ of the random measurement loss process, if $\lim_{k\to\infty}\Phi_k(X)$ exists, denote it by $P(\gamma_k|_{k=1}^{\infty})$. Using this notation, define a set $\mathcal{P}$ as

$$\mathcal{P} = \left\{ P\left(\gamma_k|_{k=1}^{\infty}\right) \middle| P\left(\gamma_k|_{k=1}^{\infty}\right) = \lim_{n\to\infty} \mathbf{H}_m\left(\prod_{i=n}^{1} M^{[\gamma_i]},\ X\right), \right. \\ \left. X > 0,\ \gamma_i \in \{0,\ 1\} \right\} \tag{7}$$

Then, Theorem 1 makes it clear that when Conditions I–III are satisfied, this matrix set is not empty and is independent of a particular PDM X. On the other hand, from its definition and (5), it is obvious that this matrix set consists of all the possible final values of the random process $P_{k|k}$. Moreover, as its element $P(\gamma_k|_{k=1}^{\infty})$ can be associated with the number $\sum_{i=1}^{\infty}\gamma_i 2^{i-1}$, the set $\mathcal{P}$ is apparently countable.

On the other hand, when Conditions I–III are satisfied, from [2, Proposition 6] and [23, Theorem 5], it can be claimed that the random process $P_{k|k}$ converges to a stationary distribution. Theorem 1 makes it clear that this stationary distribution is unique and the convergence rate is exponential. Moreover, the set $\mathcal{P}$ defined in (7) includes the support of this stationary distribution as a subset.

Under Condition II, a well established conclusion in system theory is that the algebraic Riccati equation $P = \left[\left(A^{[1]}PA^{[1]T} + G^{[1]}G^{[1]T}\right)^{-1} + H^{[1]T}H^{[1]}\right]^{-1}$ has a unique stabilizing solution. This stabilizing solution is denoted by $P^{\star}$ throughout the rest of this paper. Moreover, a widely known result in Kalman filtering is that under this condition, the Riccati recursion $P_{k+1|k+1} = \left[\left(A^{[1]}P_{k|k}A^{[1]T} + G^{[1]}G^{[1]T}\right)^{-1} + H^{[1]T}H^{[1]}\right]^{-1}$ converges to $P^{\star}$ with the increment of the temporal variable k [8], [18], which is in accordance to the recursion of (1) with $\alpha = 1$ and $\beta = 0$, i.e., there is no data dropping.

On the basis of these results, ergodicity of the random process $P_{k|k}$ is established from Lemma 2.

Corollary 1: In addition to Conditions I–III, if $P_{k|k}$ starts from $P^{\star}$, then, this random process is also ergodic.

Proof: When Conditions I–III are satisfied, the random process $P_{k|k}$ exponentially converges to a unique stationary distribution. Assume that $\gamma_k \equiv 1$. Then, for an arbitrary PDM X

$$P_{k|k} = \mathbf{H}_m\left(M^{[1]k},\ X\right), \quad k = 1, 2, \cdots \tag{8}$$

Under Condition II, from the convergence of the Kalman filter [8], [18], we have that $\lim_{k\to\infty}\mathbf{H}_m(M^{[1]k}, X)$ exists and is equal to $P^{\star}$. Moreover, from the definition of the matrix $P^{\star}$,

[1]More specifically, simultaneous satisfaction of these two conditions imply the existence of a finite positive integer m, such that the mapping $\mathbf{H}_m(M^{[1]m}, X)$ is strictly contractive. As the actual value of this integer does not affects results of this paper, it is assumed to be 1 throughout this paper.

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it is obvious that $\mathbf{H}_m(M^{[1]}, P^\star) = P^\star$. Therefore, $P^\star$ belongs to the support of the stationary distribution of the random process $P_{k|k}$.

It can therefore be declared from Lemma 2 that the random process $P_{k|k}$ initialized with $P_{0|0} = P^\star$ is ergodic. This completes the proof. $\diamond$

When both α and β belong to the open set (0, 1), it can be directly proved, as what has been done in [11], that the Markov chain $\gamma_k|_{k=1}^{\infty}$ has a stationary distribution. Denote the random variable of this stationary distribution by γ_∞. Then, the probability that γ_∞ takes the value of 1 or 0 does not depend on the temporal variable k, which can be respectively expressed as $\mathbf{P}_r(\gamma_\infty = 1) = (1-\beta)/(2-\alpha-\beta)$ and $\mathbf{P}_r(\gamma_\infty = 0) = (1-\alpha)/(2-\alpha-\beta)$.

From Corollary 1, it is clear that the stationary distribution of the random process $P_{k|k}$ can be approximated well by its time series samples. To clarify accuracy of this approximation, some properties of a Markov process are utilized that are given in Lemma 1.

For a binary series $\gamma_i|_{i=0}^{-\infty}$ with $\gamma_i \in \{0,1\}$, denote $\sum_{i=0}^{-\infty}\gamma_i 2^i$ by n and define $P^{[n]}$ as $P^{[n]} = \lim_{k\to\infty}\mathbf{H}_m(M^{[\gamma_0]}M^{[\gamma_{-1}]}\cdots M^{[\gamma_{-k}]}, P^\star)$, provided that the limit exists. Moreover, for a prescribed positive number ε, define the set $\mathcal{P}^{[n]}(\varepsilon)$ of PDMs as

$$\mathcal{P}^{[n]}(\varepsilon) = \left\{P \;\middle|\; \delta\left(P^{[n]},\ P\right) \le \varepsilon,\ P \ge 0\right\} \tag{9}$$

Then, according to Theorem 1, for any n_1 and n_2 that can be expressed as $\sum_{i=0}^{-\infty}\gamma_i 2^i$ and the corresponding $P^{[n_1]}$ and $P^{[n_2]}$ exist, there is at least one finite length binary sequence $\gamma_i^{[n_1,n_2]}|_{i=1}^{N(n_1,n_2)}$ with $\gamma_i^{[n_1,n_2]} \in \{0,1\}$, such that

$$\mathbf{H}_m\left(\prod_{i=N(n_1,n_2)}^{1} M^{\left[\gamma_i^{[n_1,n_2]}\right]}, P^{[n_1]}\right) \in \mathcal{P}^{[n_2]}(\varepsilon), \text{ in probability} \tag{10}$$

Note that when $\bar{\gamma}_i = \gamma_{i-k}$, we have that $M^{[\bar{\gamma}_i]} = M^{[\gamma_{i-k}]}$, $i = 0, 1, \ldots, k$. This means that

$$\mathbf{H}_m\left(M^{[\gamma_k]}M^{[\gamma_{k-1}]}\cdots M^{[\gamma_1]},\ P^\star\right) = \mathbf{H}_m\left(M^{[\bar{\gamma}_0]}M^{[\bar{\gamma}_{-1}]}\cdots M^{[\bar{\gamma}_{-k}]},\ P^\star\right) \tag{11}$$

and this relation is valid for all the positive integer (including $+\infty$). In addition, it can be declared from Theorem 1 that under Conditions I–III, $\lim_{k\to\infty}\mathbf{H}_m(M^{[\gamma_k]}M^{[\gamma_{k-1}]}\cdots M^{[\gamma_1]}, P^\star)$ exists in probability. Therefore, the matrix set $\mathcal{P}$ defined in Equation (7) can also be expressed as

$$\mathcal{P} = \left\{P^{[n]} \;\middle|\; P^{[n]} = \lim_{k\to-\infty}\mathbf{H}_m\left(\prod_{i=0}^{k} M^{[\gamma_i]},\ P^\star\right),\ n = \sum_{i=0}^{-\infty}\gamma_i 2^i,\ \gamma_i \in \{0,\ 1\}\right\} \tag{12}$$

In other words, the set $\mathcal{P}$ can be parametrized by $P^{[n]}$.

On the other hand, Theorem 1 declares that when Conditions I–III are satisfied, $\lim_{k\to\infty}\rho_k(X,Y) = 0$ in probability for arbitrary PDMs X and Y. It can therefore be declared that for arbitrary $P^{[p]}$ and $P^{[q]}$ belonging to the set $\mathcal{P}$, there exists a binary series $\gamma_j^{[pq]}|_{j=1}^{\infty}$ with $\gamma_j^{[pq]} \in \{0,1\}$, such that $P^{[p]} = \lim_{k\to\infty}\mathbf{H}_m(M^{[\gamma_k^{[pq]}]}M^{[\gamma_{k-1}^{[pq]}]}\cdots M^{[\gamma_1^{[pq]}]}, P^{[q]})$ is valid in probability. In addition, it has been mentioned before that for an arbitrary positive ε, only finite steps are required in probability to transform an element of $\mathcal{P}^{[p]}(\varepsilon)$ to an element of the set $\mathcal{P}^{[q]}(\varepsilon)$ by the recursions of (1). Note that $\mathcal{P}^{[p]}(\varepsilon)$ degenerates into $\{P^{[p]}\}$ when ε decreases to 0. This means that the Markov chain $P_{k|k}$ is approximately irreducible and positive recurrent.

Based on these observations, the following results are obtained, which give an approximation of the stationary distribution of the random process $P_{k|k}$ and its approximation accuracy. Their proof is deferred to the Appendix.

Theorem 2: Let $F(x)$ denote the distribution function of $\delta(P_{\infty|\infty}, P^\star)$, $P_{k|k}$ the random matrix recursion of (1) with its initial value $P_{0|0} = P^\star$ and the corresponding Markov chain $\gamma_k|_{k=0}^{\infty}$ at its stationary state. For an arbitrary positive number d, define a set $\mathcal{B}_d$ as $\mathcal{B}_d = \{P|\delta(P,P^\star) \le d\}$. Then, when Conditions I–III are satisfied

$$\lim_{n\to\infty}\frac{1}{n+1}\sum_{k=0}^{n} I_{\mathcal{B}_d}(P_{k|k}) = F(d), \quad \text{in probability} \tag{13}$$

and the convergence rate is $O((ln(n)/n)^{1/4})$.

From Theorem 2, it can be declared that when the stationary distribution of the random process $P_{k|k}$ is approximated by that of its samples, the approximation accuracy is of order $(ln(n)/n)^{1/4}$. Therefore, when a large number of the samples of $P_{k|k}$ are available, the distribution function of the stationary process can be approximated in a high accuracy.

Compared to the results of [9], Theorem 2 is valid for a random data loss process described by Markov chains which covers a larger class of engineering problems. In addition, it provides not only an approximation for the stationary distribution of the random process $P_{k|k}$, but also an order of approximation accuracy.

IV. Approximation of the Stationary Distribution

In the previous section, it has been proved that when $P_{k|k}$ starts from $P^\star$ and the Markov chain γ_k is in its stationary state, the corresponding random sequence $P_{k|k}$ is ergodic. These results make it possible to approximate the stationary distribution of $P_{k|k}$ using its samples. In this section, some explicit formulas are given for approximations of this stationary distribution in which actual sampling on all $P_{k|k}$ is not required.

To investigate this approximation, the following results are at first established, which makes it clear that in finite recursions, the Homographic transformation of (4) generally can not remove influences of its initial values.

Lemma 4: Assume that Condition I is satisfied. Then, for arbitrary PDMs X and Y with a compatible dimension, $\mathbf{H}_m(M^{[*]}, X) = \mathbf{H}_m(M^{[*]}, Y)$ if and only if $X = Y$, no matter whether $* = 1$ or $* = 0$.

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Proof: From the definition of the Homographic transformation, direct algebraic manipulations show that when X and Y are positive definite and their dimensions are compatible

$$\begin{aligned}&\mathbf{H}_m\left(M^{[0]},X\right)-\mathbf{H}_m\left(M^{[0]},Y\right)\\&=\left[A^{[0]}XA^{[0]T}+G^{[0]}G^{[0]T}\right]-\left[A^{[0]}YA^{[0]T}+G^{[0]}G^{[0]T}\right]\\&=A^{[0]}(X-Y)A^{[0]T}\end{aligned}\tag{14}$$

$$\begin{aligned}&\mathbf{H}_m\left(M^{[1]},X\right)-\mathbf{H}_m\left(M^{[1]},Y\right)\\&=\left[\left(A^{[1]}XA^{[1]T}+G^{[1]}G^{[1]T}\right)^{-1}+H^{[1]T}H^{[1]}\right]^{-1}\\&\quad-\left[\left(A^{[1]}YA^{[1]T}+G^{[1]}G^{[1]T}\right)^{-1}+H^{[1]T}H^{[1]}\right]^{-1}\\&=\left[I+\left(A^{[1]}XA^{[1]T}+G^{[1]}G^{[1]T}\right)H^{[1]T}H^{[1]}\right]^{-1}A^{[1]}(X-Y)\\&\quad\times A^{[1]T}\left[I+H^{[1]T}H^{[1]}\left(A^{[1]}YA^{[1]T}+G^{[1]}G^{[1]T}\right)\right]^{-1}\end{aligned}\tag{15}$$

The conclusions are immediate from these relations and the regularity of both $A^{[0]}$ and $A^{[1]}$. This completes the proof. ◇

Assume that the Markov chain γ_k is in its stationary state, and $P_{k|k}$ starts from $P^\star$. Let $P_{0|0},P_{1|1},\ldots,P_{n|n}$ be its first $n+1$ samples, and consider all the possible values that these samples may take and the probability of their occurrence. Obviously from Lemma 4, when Condition I is satisfied, there are 2^k possible values that $P_{k|k}$ may take, which is in accordance with all the realizations of the Markov chain $\gamma_i|_{i=1}^k$ with $\gamma_i\in\{0,1\}$. Recall that $\mathbf{H}_m(M^{[1]},P^\star)=P^\star$. It is clear that for every positive integer k

$$\begin{aligned}\mathbf{H}_m\left(M^{[1]k},\ P^\star\right)&=\mathbf{H}_m\left[M^{[1](k-1)},\mathbf{H}_m\left(M^{[1]},\ P^\star\right)\right]\\&=\mathbf{H}_m\left(M^{[1](k-1)},\ P^\star\right)=\cdots\\&=P^\star\end{aligned}\tag{16}$$

On the other hand, if it exists, let $P^\S$ denote the solution of the algebraic Lyapunov equation $P=A^{[0]}PA^{[0]T}+G^{[0]}G^{[0]T}$ that is positive definite. This happens when the matrix $A^{[0]}$ is stable [8], [18]. From Lemma 4, it is clear that if $P\notin\{P^\star,P^\S\}$, then, $\mathbf{H}_m(M^{[*]},P)\neq P$, no matter whether $*$ is equal to 1 or 0.

From these arguments, the following results can be obtained, while their proof is included in the Appendix.

Lemma 5: Let $\bar{\mathcal{P}}^{[n]}$ denote the set consisting of all possible values that $P_{k|k}|_{k=0}^n$ may take which has its initial value being $P^\star$ and recursively updates according to the stationary process of the Markov chain γ_k. Then, the number of the elements in $\bar{\mathcal{P}}^{[n]}$ is equal to 2^n and the set $\bar{\mathcal{P}}^{[n]}$ can be expressed as

$$\begin{aligned}\bar{\mathcal{P}}^{[n]}=&\left\{P^\star,\ \mathbf{H}_m\left(M^{[0]},\ P^\star\right)\right\}\\&\bigcup\left\{P\ \middle|\ \begin{array}{c}P=\mathbf{H}_m\left[M^{[\gamma_k]}M^{[\gamma_{k-1}]}\cdots M^{[\gamma_2]},\mathbf{H}_m\left(M^{[0]},\ P^\star\right)\right]\\ \gamma_j\in\{0,1\},j\in\{2,3,\ldots,k\},k\in\{2,3,\ldots,n\}\end{array}\right\}\end{aligned}\tag{17}$$

For any sequence $\gamma_j|_{j=1}^k$ with $\gamma_j\in\{0,1\}$ and $k\in\{1,2,\ldots,n-1\}$, define $l(\gamma_j|_{j=1}^k)$ and $\bar{P}^{[l(\gamma_j|_{j=1}^k)]}$ respectively as $l(\gamma_j|_{j=1}^k)=1+2^{k-1}+\sum_{j=1}^{k-1}\gamma_j2^{j-1}$ and $\bar{P}^{[l(\gamma_j|_{j=1}^k)]}=\mathbf{H}_m(M^{[\gamma_k]}M^{[\gamma_{k-1}]}\cdots M^{[\gamma_1]}M^{[0]},P^\star)$. Moreover, define $\bar{P}^{[1]}=P^\star$. Clearly, $l(\gamma_j|_{j=1}^k)$ provides a natural order for each element of the set $\bar{\mathcal{P}}^{[n]}$. From the proof of Lemma 5, it can be understood that $\mathcal{P}^{[n]}=\{\bar{P}^{[1]},\bar{P}^{[2]},\ldots,\bar{P}^{[2^n]}\}$.

The following theorem gives the convergence value of $(1/(n+1))\sum_{k=0}^n I_{\mathcal{B}_d}(P_{k|k})$, which is helpful in deriving approximations for the stationary distribution of $P_{k|k}$. Its proof is given in the Appendix.

Theorem 3: For each prescribed positive d, define a set $\mathcal{N}_d$ as $\mathcal{N}_d=\{j|\bar{P}^{[j]}\in\bar{\mathcal{P}}^{[n]}\bigcap\mathcal{B}_d\}$. Moreover, denote $(1-\beta)/(2-\alpha-\beta)$ by γ_{st}. Then

$$\begin{aligned}\lim_{n\to\infty}\frac{1}{n+1}\sum_{k=0}^n I_{\mathcal{B}_d}(P_{k|k})=&\lim_{n\to\infty}\sum_{j\in\mathcal{N}_d}\left(1-\gamma_{st}^{n-\lceil log_2(j)\rceil}\right)\\&\times\gamma_{st}^{\sum_{i=1}^{\lceil log_2(j)\rceil}\gamma_i^{[j]}}(1-\gamma_{st})^{\lceil log_2(j)\rceil-\sum_{i=1}^{\lceil log_2(j)\rceil}\gamma_i^{[j]}}\end{aligned}\tag{18}$$

in which $\gamma_i^{[j]}$ is the binary code for $j-1-2^{\lceil log_2(j)\rceil-1}$.

In the above theorem, an explicit formula is given for the stationary distribution of the random process $P_{k|k}$. In principle, its value can be computed for each prescribed d, which means that this distribution function can be obtained to an arbitrary accuracy, provided that a computer is available which has a sufficient computation speed and a sufficient memory capacity. Note that the value of 2^n increases exponentially with the increment of the sample size n, and a large n is generally appreciated as it leads to a more accurate approximation. It appears reasonable to claim that in general, conclusions of the above theorem can not be directly utilized in actual computations, and some other computationally more efficient approximations are still required.

From (18), however, it is obvious that when γ_{st} is approximately equal to 1, any element $P^{[j]}$ of the set $\bar{\mathcal{P}}^{[n]}$ with a large $\lceil log_2(j)\rceil-\sum_{i=1}^{\lceil log_2(j)\rceil}\gamma_i^{[j]}$ has a small probability to occur. Note that this number is in fact equal to that of the zeros in the associated binary sequence. This result has some nice physical interpretations, as these samples are associated with many data transmission failures which can hardly happen if $\gamma_{st}\approx 1$. On the other hand, from the proof of Theorem 1, it is clear that with the increment of the length of the binary sequence, which is equivalent to the increment of the number j, the probability that it has a large number of zeros also increases. These mean that contributions of an element $P^{[j]}\in\bar{\mathcal{P}}^{[n]}$ with a large j to the stationary distribution of the random process $P_{k|k}$ are usually very small and therefore negligible.

On the other hand, note that $\mathbf{H}_m(M^{[0]},X)=A^{[0]}XA^{[0]T}+G^{[0]}G^{[0]T}$. It is straightforward to prove from the definition of the Riemannian distance that for an arbitrary PDM X, there exist finite positive numbers a and b that do not depend on the matrix X, such that

$$\delta\left[\mathbf{H}_m\left(M^{[0]},X\right),\ P^\star\right]\leq a\delta(X,P^\star)+b\tag{19}$$

On the basis of this inequality, the following property is established for the matrix set $\bar{\mathcal{P}}^{[n]}$. Its proof is deferred to the Appendix.

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Theorem 4: For a prescribed positive integer n (including $+\infty$), define a set $\tilde{\mathcal{P}}^{[n]}$ as $\tilde{\mathcal{P}}^{[n]} = \{P | P = \mathbf{H}_m(M^{[0]i}, P^\star), i = 0, 1, \ldots, n\}$. Then, when Conditions I–III are satisfied

$$\lim_{\alpha_{1h}\to 0} \sup_{P\in\tilde{\mathcal{P}}^{[n]}} \inf_{Q\in\tilde{\mathcal{P}}^{[n]}} \delta(P, Q) = 0 \tag{20}$$

In addition, it is obvious from the definition of the set $\tilde{\mathcal{P}}^{[n]}$ that if $P_{0|0} = P^\star$, then, $P_{k|k} \notin \tilde{\mathcal{P}}^{[n]}$ only if $k \geq 2$ and there are more than two temporal instants at which the communication channel fails. Hence, it can be directly proved from (A41) that for every integer n greater than 2

$$\mathbf{P}_r\left\{\sup_{P\in\tilde{\mathcal{P}}^{[n]}} \inf_{Q\in\tilde{\mathcal{P}}^{[n]}} \delta(P, Q) > 0\right\} \leq 1-\gamma_{st}^{n-1}(n+\gamma_{st}-n\gamma_{st}) \tag{21}$$

From this inequality and Theorems 3 and 4, it can be declared that when α_{1h} is small and/or γ_{st} is close to 1, there certainly exits a finite integer n such that with a high probability, all the samples of the random process $P_{k|k}$ in its stationary state are concentrated around the elements of the set $\tilde{\mathcal{P}}^{[n]}$, and a matrix far away from this set usually has a negligible probability to occur. These concentrations become more dominating if both a and b are not very large and α_{1h} is significantly smaller than 1, which can be understood from (A46)–(A50).

From these observations, it seems reasonable to approximate the support of $P_{\infty|\infty}$ by the set $\tilde{\mathcal{P}}^{[\infty]}$. When this approximation is valid, a very simple and explicit formula can be derived for the stationary distribution of the random process $P_{k|k}$, which is given in the next theorem. Its proof is deferred to the Appendix.

Theorem 5: Assume that α_{1h} is sufficiently small and/or γ_{st} is sufficiently close to 1, such that the set $\tilde{\mathcal{P}}^{[\infty]}$ is a good approximation for the support of the stationary process of $P_{k|k}$. Then

$$\mathbf{P}_r\left\{P_{\infty|\infty} = \mathbf{H}_m\left(M^{[0]i}, P^\star\right)\right\} \approx \gamma_{st}(1-\gamma_{st})^i,\ i=0,1,2,\ldots \tag{22}$$

Note that when $\gamma_{st} \approx 1$, $\gamma_{st}(1-\gamma_{st})^i$ decreases rapidly to 0 with the increment of the index i. This means that when the data arrival probability is high, only a few elements of the set $\mathcal{P}^{[\infty]}$, that is, $\mathbf{H}_m(M^{[0]i}, P^\star)$, are required in computing the approximation for the stationary distribution of the random process $P_{k|k}$. Another attractive characteristic of this approximation is that its accuracy does not depend on the length of time series samples, and therefore can greatly reduce computation burdens.

Note also that the Kalman filter with data loss never outperforms that without data loss. This means that $P_{\infty|\infty} - P^\star \geq 0$ is always valid [9], [12], [19], [23]. As practical communication channels usually do not have a high data loss probability, it can be declared from Theorem 5 that $\delta(\mathbf{H}_m(M^{[0]}, P^\star), P^\star)$ is a good index for steady performance deterioration due to data loss. Moreover, to reduce influence of data loss on state estimation accuracy, in addition to improve communication qualities, another possible method is to design a plant that makes this index as small as possible, which may be achieved through measurement variable selection, etc.

When random data loss is described by a Bernoulli process and the data arrival probability is approximately equal to 1, it is suggested in [9] to approximate the stationary distribution of the CMEE of the Kalman filter with a delta function concentrated on $P^\star$. While this approximation is intuitively understandable, its approximation accuracy is still not clear, as there are still no computationally efficient methods for determining parameters of the required rate function. On the other hand, the results of Theorems 4 and 5 suggest that rather than a delta function, it is more appropriate to approximate this stationary distribution with a series of delta functions. This difference has some significant influences on steady update gain matrix selection of the state estimator.

More precisely, a well known fact about Kalman filter is that in order to reduce computational complexity, its update gain matrix can be replaced by the steady value without any influences on its steady estimation accuracy. Note that in both the Kalman filter and the robust state estimator of [23], the update gain matrix is equal to $P_{k+1|k+1}C_{k+1}^T(0)R_{k+1}^{-1}$ when a measurement arrives. Here, $C_k(0)$ and R_k stand respectively for the nominal value of the plant output matrix and the covariance matrix of measurement errors at the time instant k. This means that when the nominal plant is time invariant, distribution of the update gain matrix of the estimator is completely determined by that of $P_{k+1|k+1}$. Therefore, the results of [9] suggests that when the measurement arrival probability is close to 1, the update gain matrix of the Kalman filter can be replaced by $P^\star C_{k+1}^T(0)R_{k+1}^{-1}$ without significant sacrifice of steady estimation accuracy, which is consistent with the aforementioned conclusions about conventional Kalman filtering. Theorem 5, however, suggests that this simple update gain matrix replacement generally deteriorates steady estimation accuracy, no matter whether the Kalman filter or the robust state estimator of [23] is utilized. This deterioration usually becomes more significant with the increment of the distance between $P^\star$ and $\mathbf{H}_m(M^{[0]}, P^\star)$.

While Theorem 5 provides a very simple approximation and our simulations show that this approximation is usually very accurate, it is still a challenging problem to derive a tighter bound on its approximation errors, as well as more explicit conditions on system parameters under which this approximation is valid.

V. A Numerical Example

To illustrate accuracies of the derived approximations, various numerical simulations have been performed. Some typical results are reported in this section. The adopted plant has a state space model $x_{k+1} = Ax_k + Gw_k$, $y_k = Hx_k + v_k$, with

$$A = \begin{bmatrix} -9.3259\times10^{-1} & 3.1968\times10^{-1} & 5.4623\times10^{-1} \\ -6.0799\times10^{-1} & 1.7753\times10^{-1} & -5.6542\times10^{-1} \\ 7.4219\times10^{-1} & 4.7183\times10^{-1} & 5.0898\times10^{-1} \end{bmatrix}$$

$$G = \begin{bmatrix} -9.8569\times10^{-1} & 3.1333\times10^{-1} \\ 7.4412\times10^{-1} & -1.0234\times10^{-1} \\ -5.5144\times10^{-1} & -8.4089\times10^{-1} \end{bmatrix}$$

$$H = \begin{bmatrix} 5.2282\times10^{-1} & 1.3088\times10^{-1} & -2.1592\times10^{-1} \\ -6.3811\times10^{-1} & -1.8414\times10^{-1} & 4.9641\times10^{-1} \end{bmatrix}$$

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Moreover, w_k and v_k are independent zero-mean Gaussian white noises with $\mathbf{V}_{ar}\{w_k\} \equiv I_2$ and $\mathbf{V}_{ar}\{v_k\} \equiv 1$.

This model is randomly selected from the set of unstable, but controllable and observable systems with each of its parameters independently and uniformly distributed over the interval $[-1, 1]$.

Direct numerical computations show that the eigenvalues of the matrix A are -1.1395, $4.4670 \times 10^{-1} \pm j5.6951 \times 10^{-1}$, while the singular values of its controllability matrix and observability matrix are respectively 2.1852, 1.6303, 8.7564×10^{-1} and 2.0984, 5.4447×10^{-1}, 4.0892×10^{-1}. These results confirm that this plant is indeed unstable, controllable and observable. On the other hand, using the Matlab command *dric.m*, the following $P^\star$ is obtained

$$P^\star = \begin{bmatrix} 1.1962 & -6.1302\times10^{-1} & 6.2078\times10^{-1} \\ -6.1302\times10^{-1} & 1.8190 & -8.0186\times10^{-1} \\ 6.2078\times10^{-1} & -8.0186\times10^{-1} & 1.4136 \end{bmatrix}$$

Note that the plant adopted in [23] is stable. It can be claimed that the above simulation system is more appropriate in investigating typical stationary behaviors of a state estimator with random data loss, recalling that if the system is stable, both the CMEE of the Kalman filter and the PCM of the robust state estimator of [23] converges to a constant PDM with the increment of the temporal variable k, even when all the measured data are lost [12], [16], [19], [23].

It is worthwhile to mention that the numerical example of [23] is used to illustrate that its robust estimation procedure is more accurate than the Kalman filter when modelling errors exist in a plant model, and its PCM converges when some weak conditions are satisfied. The numerical example of this section, however, is to demonstrate the approximation accuracy of Theorems 2 and 5 on the stationary distribution of the CMEE of the Kalman filter.

Various situations have been tested on this numerical example. The obtained computation results confirm the theoretical results established in the previous sections. In these simulations, both empirical stationary distribution of the random CMEE process and the approximations based respectively on its ergodicity property and the delta functions are computed. In computing the empirical stationary distribution, 5×10^4 trials are performed for simulating the CMEE at $k = 10^3$ that are initialized with $\mathbf{P}_r(\gamma_0 = 1) = 0.7$ and a PDM $P_{0|0} = 10^3 I_2$, and the empirical stationary distribution is calculated using the obtained $\delta(P_{10^3|10^3}, P^\star)$. When the approximation of Theorem 2 is used, the CMEE $P_{k|k}$ is initialized with $P_{0|0} = P^\star$ and $\mathbf{P}_r(\gamma_0 = 1) = (1-\beta)/(2-\alpha-\beta)$ which is the stationary distribution of the Markov chain γ_k. The first 5×10^4 samples of the CMEE, that is, $P_{k|k}|_{k=1}^{5\times10^4}$, are simulated which are further used to compute an approximation of the stationary distribution of the CMEE on the basis of Theorem 2.

In computing the empirical stationary distribution and its Theorem 2 based approximation, an interval $[0, \delta_{\max}]$ is at first divided into N_e intervals of an equal length, in which N_e and $\delta_{\max}$ are some prescribed positive integer and number that are suitably selected according to the maximal value of the distance from the simulated CMEEs to the matrix $P^\star$. Then, the number of the simulated CMEE samples are counted that belong to $[i(\delta_{\max}/N_e), (i+1)(\delta_{\max}/N_e))$, $i = 0, 1, \ldots, N_e - 1$. This number is divided by the total number of the simulated samples, and is regarded to be a value proportional to that of the empirical probability density function (PDF) of the stationary CMEE process or its Theorem 2 based approximation at $\delta = (i + (1/2))(\delta_{\max}/N_e)$. Moreover, the points determined by these values are connected using the Matlab command *plot.m*. The obtained curves are regarded to be proportional to those of the empirical PDF and its approximation using Theorem 2 (the proportional rate is $\delta_{\max}/N_e$.). To make statements concise, with a little abuse of terminology, these curves are respectively called empirical PDF and its approximation.

When the approximation of Theorem 5 is used, the following method is adopted for comparing results of empirical distributions of the stationary CMEE process and its approximations. At first, select a suitable positive integer N_d, and compute $d_i = \delta(\mathbf{H}_m(M^{[0]i}, P^\star), P^\star)$ and $p_i = \gamma_{st}(1-\gamma_{st})^i$, $i = 1, 2, \ldots, N_d$. This N_d is chosen to guarantee that $\gamma_{st}(1-\gamma_{st})^{N_d+1}$ is smaller than some threshold value, for example, 10^{-7}. Then, another positive integer N_s is selected according to the distance distribution between the simulated CMEE and the matrix $P^\star$, which is used to reflect the closeness of the simulated stationary distribution to delta functions. Afterwards, the number is counted of the simulated CMEEs that have a distance to the matrix $P^\star$ belonging to the interval $\mathcal{I}_i$, $i = 0, 1, \ldots, N_d$, in which

$$\mathcal{I}_i = \begin{cases} \left[0,\ \frac{d_1}{N_s}\right], & i = 0 \\ \left(d_i - \frac{d_i - d_{i-1}}{N_s},\ d_i + \frac{d_{i+1} - d_i}{N_s}\right], & i = 1, 2, \ldots, N_d - 1 \\ \left(d_{N_d} - \frac{d_{N_d} - d_{N_d-1}}{N_s},\ d_{N_d}\right], & i = N_d \end{cases}$$

Finally, these numbers are divided by the total number of the simulated samples, and regarded to be the empirical value of the probability of the stationary CMEE process and its Theorem 2 based approximation at $\delta(P_{\infty|\infty}, P^\star) = d_i$. Clearly, under the condition that these values are close to p_i, the greater the integer N_s is, the closer the stationary distribution of the random CMEE process to the delta functions.

The left column of Fig. 1 plots the empirical value of PDF of the stationary CMEE process and its Theorem 2 based approximation, while the corresponding empirical value of the probability is plotted in the right column, together with its approximations based respectively on Theorems 2 and 5.

In Fig. 1(a) and (b), simulations results with $\alpha = 0.95$ and $\beta = 0.05$ are given, in which $\delta_{\max}$, N_e, N_d and N_s are respectively chosen as $\delta_{\max} = 1.4$, $N_e = 2.60 \times 10^2$, $N_d = 5$ and $N_s = 4$. To understand the approximation accuracy more clearly, the computed values used in plotting Fig. 1(b) are provided in Table I.

From these results, it is clear that when the data loss probability is low, which corresponds to a large α and a small β, the PDF of the stationary CMEE process is really very close to a series of delta functions, and the approximation based on either Theorem 2 or Theorem 5 has a high accuracy.

When $\alpha = 0.80$ and $\beta = 0.30$, the corresponding simulated results are given in Table II and Fig. 1(c) and (d). In the

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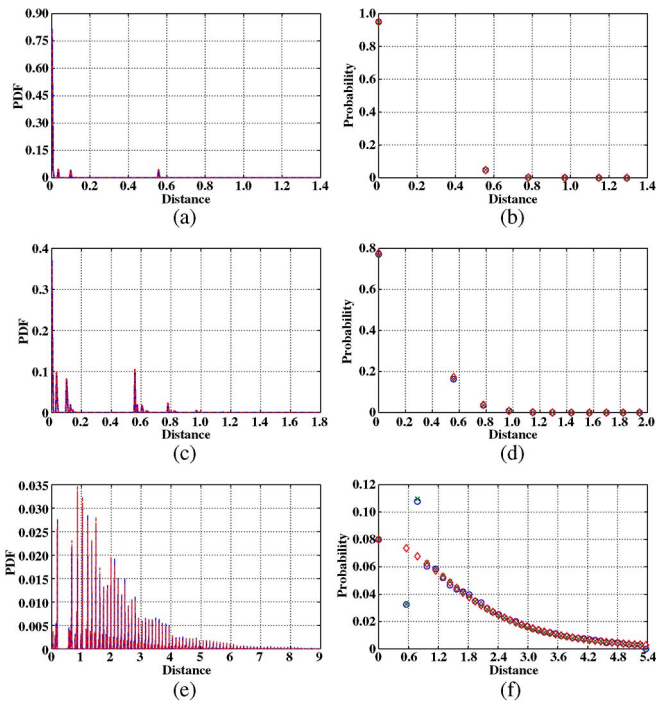

Fig. 1. Empirical PDF and probability of the stationary CMEE process, together with their approximations. ——: empirical PDF; — · —: PDF approximation based on Theorem 2; ○: empirical probability; X: probability approximation based on Theorem 2; ◇: probability approximation based on Theorem 5. (a) PDF ($\alpha = 0.95$, $\beta = 0.05$); (b) probability ($\alpha = 0.95$, $\beta = 0.05$); (c) PDF ($\alpha = 0.80$, $\beta = 0.30$); (d) probability ($\alpha = 0.80$, $\beta = 0.30$); (e) PDF ($\alpha = 0.08$, $\beta = 0.92$); (f) probability ($\alpha = 0.08$, $\beta = 0.92$).

TABLE I
EMPIRICAL PROBABILITY AND ITS APPROXIMATIONS
($\alpha = 0.95$, $\beta = 0.05$)

Distance (d_i)	App.Theorem 5	App.Theorem 2	Empirical Prob.
0	9.5000×10^{-1}	9.4738×10^{-1}	9.4982×10^{-1}
5.5743×10^{-1}	4.7500×10^{-2}	4.9280×10^{-2}	4.7500×10^{-2}
7.7934×10^{-1}	2.3750×10^{-3}	2.6800×10^{-3}	2.3800×10^{-3}
9.6938×10^{-1}	1.1875×10^{-4}	2.2000×10^{-4}	1.0000×10^{-4}
1.1470	5.9375×10^{-6}	0	0
1.2928	2.9688×10^{-7}	0	0

TABLE II
EMPIRICAL PROBABILITY AND ITS APPROXIMATIONS
($\alpha = 0.80$, $\beta = 0.30$)

Distance (d_i)	App.Theorem 5	App.Theorem 2	Empirical Prob.
0	7.7778×10^{-1}	7.6832×10^{-1}	7.7078×10^{-1}
5.5743×10^{-1}	1.7284×10^{-1}	1.6592×10^{-1}	1.6290×10^{-1}
7.7934×10^{-1}	3.8409×10^{-2}	3.5560×10^{-2}	3.6000×10^{-2}
9.6938×10^{-1}	8.5353×10^{-3}	8.2000×10^{-3}	8.8200×10^{-3}
1.1470	1.8967×10^{-3}	1.7400×10^{-3}	1.5000×10^{-3}
1.2928	4.2150×10^{-4}	4.2000×10^{-4}	4.0000×10^{-4}
1.4335	9.3666×10^{-5}	1.4000×10^{-4}	4.0000×10^{-5}
1.5679	2.0815×10^{-5}	4.0000×10^{-5}	0
1.6945	4.6255×10^{-6}	2.0000×10^{-5}	0
1.8191	1.0279×10^{-6}	0	0
1.9408	2.2842×10^{-7}	0	0

related computations, $\delta_{\max} = 1.8$, $N_e = 3.50 \times 10^2$, $N_d = 10$ and $N_s = 4$ are utilized. These results show that when the data loss probability has a moderate value, the suggested approximations are still of a high accuracy.

TABLE III
EMPIRICAL PROBABILITY AND ITS APPROXIMATIONS
($\alpha = 0.08$, $\beta = 0.92$)

Distance (d_i)	App.Theorem 5	App.Theorem 2	Empirical Prob.
0	8.0000×10^{-2}	8.0480×10^{-2}	7.9920×10^{-2}
5.5743×10^{-1}	7.3600×10^{-2}	3.2720×10^{-2}	3.2480×10^{-2}
7.7934×10^{-1}	6.7712×10^{-2}	1.0964×10^{-1}	1.0778×10^{-1}
9.6938×10^{-1}	6.2295×10^{-2}	6.3000×10^{-2}	6.0380×10^{-2}
1.1470	5.7311×10^{-2}	5.8040×10^{-2}	5.8580×10^{-2}
1.2928	5.2727×10^{-2}	5.3480×10^{-2}	5.1920×10^{-2}
1.4335	4.8508×10^{-2}	4.9280×10^{-2}	4.6640×10^{-2}
1.5679	4.4628×10^{-2}	4.5140×10^{-2}	4.3580×10^{-2}
1.6945	4.1058×10^{-2}	4.1300×10^{-2}	4.1820×10^{-2}
1.8191	3.7773×10^{-2}	3.8360×10^{-2}	3.9760×10^{-2}
1.9408	3.4751×10^{-2}	3.4800×10^{-2}	3.4880×10^{-2}
2.0601	3.1971×10^{-2}	3.1740×10^{-2}	3.3800×10^{-2}
2.1783	2.9413×10^{-2}	2.8960×10^{-2}	2.9580×10^{-2}
2.2952	2.7060×10^{-2}	2.6600×10^{-2}	2.6880×10^{-2}
2.4113	2.4895×10^{-2}	2.4480×10^{-2}	2.5140×10^{-2}
2.5268	2.2904×10^{-2}	2.2760×10^{-2}	2.2820×10^{-2}
2.6418	2.1071×10^{-2}	2.0620×10^{-2}	2.0960×10^{-2}
2.7564	1.9386×10^{-2}	1.8820×10^{-2}	2.0160×10^{-2}
2.8708	1.7835×10^{-2}	1.7200×10^{-2}	1.7540×10^{-2}
2.9849	1.6408×10^{-2}	1.5580×10^{-2}	1.6540×10^{-2}
3.0988	1.5095×10^{-2}	1.4220×10^{-2}	1.4760×10^{-2}
3.2126	1.3888×10^{-2}	1.3120×10^{-2}	1.3740×10^{-2}
3.3264	1.2777×10^{-2}	1.2040×10^{-2}	1.2200×10^{-2}
3.4400	1.1755×10^{-2}	1.1000×10^{-2}	1.1940×10^{-2}
3.5536	1.0814×10^{-2}	1.0440×10^{-2}	1.1160×10^{-2}

Our experiences show that even when the measured data has a high probability to be lost, which corresponds to a small α and a large β, the approximation of Theorem 5 may still have a good accuracy. Fig. 1(e) and (f), as well as Table III, give some simulation results with $\alpha = 0.08$ and $\beta = 0.92$, in which $\delta_{\max} = 18$, $N_e = 3.500 \times 10^3$, $N_d = 40$ and $N_s = 2$ are utilized.[2] Clearly, the approximation of Theorem 2 still has a value close to the empirical distributions of the stationary CMEE process, but approximation errors of Theorem 5 become greater, especially at d_1 and d_2. This can be seen from Fig. 1(e), which indicates that around these two distances, the empirical PDF is no longer well approximated by a delta function. In addition, except distances near d_0, separations among successive delta functions become short, and the width of each delta function increases. All these factors affect the approximation accuracy of Theorem 5. Despite these influences, it appears that the approximation accuracy is still acceptable.

Consistent results have been obtained when the simulation settings such as the initial probability of the Markov chain, initial value of the CMEE, number of the simulated CMEEs at its stationary state, etc., are changed to other values. These results suggest that both Theorem 2 and Theorem 5 can in

[2]Due to space considerations, only results with $d_i \leq 3.5536$ are given in this table. It is clear from Fig. 1(f) that for these d_is, large differences may exist among the empirical value and the approximations. On the other hand, when $d_i > 9$, the value of the PDF is very small. To make characteristics of the curves clear, only values with $0 \leq d_i \leq 9$ are plotted in Fig. 1(e).

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general provide a highly accurate approximation for the distribution of the stationary CMEE process. Moreover, while the approximation accuracy of Theorem 5 is influenced by the parameters α and β of the Markov chain γ_k, they do not affect that of Theorem 2. But to reach a high accuracy, the approximation of Theorem 2 usually asks for a large number of time series samples.

These simulation results also make it clear that for the adopted numerical example, when $1 > \alpha \geq 0.95$ and $0.05 \geq \beta > 0$, state estimation performance is not significantly influenced by the communication channel, and the update gain matrix of the Kalman filter can be replaced by its steady value determined by the PDM $P^{\star}$. However, when $0.80 \geq \alpha > 0$ and $1 > \beta \geq 0.30$, the communication channel may greatly deteriorate state estimation accuracy, and replacement of the Kalman filter by a constant gain observer will further worsen this performance.

When the robust state estimator of [22], [23] is used, similar phenomena have been observed. That is, both Theorems 2 and 5 usually provide a good approximation to the stationary distribution of its PCM when the data loss process is a Markov chain. The detailed computation results are omitted due to space considerations.

VI. Conclusion

In this paper, asymptotic properties are investigated for both the CMEE of the Kalman filter and the PCM of a robust recursive state estimator, under the situation that the data loss process is described by a Markov chain. It has been made clear that when the associated plant is both controllable and observable, this random matrix process converges exponentially to a stationary process that does not depend on its initial value. Moreover, when these state estimators start from the stabilizing solution of the algebraic Riccati equation defined by the system parameters of the associated plant, it is shown that this random matrix process becomes ergodic. An important observation is that when the data arrival probability is approximately equal to 1, the distribution of the corresponding stationary process can be well approximated by a set of delta functions. Based on these results, two approximations have been derived for their stationary distributions, together with an error bound for one of these two approximations. It has been made clear that replacement of the update gain matrix of these estimators by a constant matrix usually sacrifice steady estimation accuracy, even if the measurement arrival probability is close to 1. Numerical simulations show that these approximations usually have a high accuracy.

As a further research, it is important to investigate characteristics of the delta functions utilized in the aforementioned approximations, as well as tighter error bounds for these approximations.

Appendix
Proof of Some Technical Results

Proof of Theorem 1: Let $*$ be any element of the set $\{0, 1\}$. It can be declared from the definitions of α_{0h} and α_{1h} that for every PDM pair X and Y, $\delta(\mathbf{H}_m(M^{[*]}, X), \mathbf{H}_m(M^{[*]}, Y)) \leq \alpha_{1h}^{*}\alpha_{0h}^{1-*}\delta(X, Y)$. Hence, for arbitrary PDMs X and Y

$$
\begin{aligned}
&\rho_k(X,Y)\\
&= \delta\left\{\mathbf{H}_m\left(\prod_{i=k}^{1} M^{[\gamma_i]}, X\right), \mathbf{H}_m\left(\prod_{i=k}^{1} M^{[\gamma_i]}, Y\right)\right\}\\
&= \delta\left\{\mathbf{H}_m\left[M^{[\gamma_k]},\ \mathbf{H}_m\left(\prod_{i=k-1}^{1} M^{[\gamma_i]},\ X\right)\right],\right.\\
&\qquad\left.\mathbf{H}_m\left[M^{[\gamma_k]},\ \mathbf{H}_m\left(\prod_{i=k-1}^{1} M^{[\gamma_i]},\ Y\right)\right]\right\}\\
&\leq \alpha_{1h}^{\gamma_k}\alpha_{0h}^{1-\gamma_k}\delta\left\{\mathbf{H}_m\left(\prod_{i=k-1}^{1} M^{[\gamma_i]}, X\right),\right.\\
&\qquad\left.\mathbf{H}_m\left(\prod_{i=k-1}^{1} M^{[\gamma_i]}, Y\right)\right\}\\
&= \alpha_{1h}^{\gamma_k}\alpha_{0h}^{1-\gamma_k}\delta\left\{\mathbf{H}_m\left[M^{[\gamma_{k-1}]},\ \mathbf{H}_m\left(\prod_{i=k-2}^{1} M^{[\gamma_i]},\ X\right)\right],\right.\\
&\qquad\left.\mathbf{H}_m\left[M^{[\gamma_{k-1}]},\ \mathbf{H}_m\left(\prod_{i=k-2}^{1} M^{[\gamma_i]},\ Y\right)\right]\right\}\\
&\leq \alpha_{1h}^{\gamma_k}\alpha_{0h}^{1-\gamma_k}\alpha_{1h}^{\gamma_{k-1}}\alpha_{0h}^{1-\gamma_{k-1}}\delta\left\{\mathbf{H}_m\left(\prod_{i=k-2}^{1} M^{[\gamma_i]},\ X\right),\right.\\
&\qquad\left.\mathbf{H}_m\left(\prod_{i=k-2}^{1} M^{[\gamma_i]},\ Y\right)\right\}\\
&= \cdots \leq \left(\prod_{i=k}^{1}\alpha_{1h}^{\gamma_i}\alpha_{0h}^{1-\gamma_i}\right)\delta(X,Y)\\
&= \alpha_{1h}^{\sum_{i=1}^{k}\gamma_i}\alpha_{0h}^{k-\sum_{i=1}^{k}\gamma_i}\delta(X,Y) \leq \alpha_{1h}^{\sum_{i=1}^{k}\gamma_i}\delta(X,Y)
\end{aligned}
\tag{A1}
$$

Define a function $f(\cdot)$ on the random process γ_k as $f(\gamma_k) = \gamma_k$. When both α and β belong to (0, 1), it is obvious that the Markov chain γ_k is positive recurrent and only has two states, that is $\gamma_k = 1$ and $\gamma_k = 0$. Using the same symbols of Lemma 1, it is obvious that for an arbitrary $j \in \{0, 1\}$

$$
f_{\alpha}^{[j]} = \sum_{k=\tau_{\alpha}^{[j]}}^{\tau_{\alpha+1}^{[j]}-1} f(\gamma_k) \leq \left(\tau_{\alpha+1}^{[j]} - 1\right) - \left(\tau_{\alpha}^{[j]} - 1\right) = \tau_{\alpha+1}^{[j]} - \tau_{\alpha}^{[j]}
\tag{A2}
$$

From this relation and properties of Markov chains, it is straightforward to prove that

$$
s(f) = \frac{1}{\mu_1} > 0,\ \mathbf{E}\left(|f_{\alpha}^{[j]}|^3\right) \leq \mathbf{E}\left(\left|\tau_{\alpha+1}^{[j]} - \tau_{\alpha}^{[j]}\right|^3\right) < \infty
\tag{A3}
$$

$$
\mathbf{V}_{ar}\left\{f_{\alpha}^{[j]} - s(f)\left(\tau_{\alpha+1}^{[j]} - \tau_{\alpha}^{[j]}\right)\right\} > 0
\tag{A4}
$$

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Moreover, both μ_1 and π_1 are positive constants. Hence, according to Lemma 1, we have that

$$\sup_{t\in\mathcal{R}}\left|\mathbf{P}_r\left\{\frac{\sqrt{\mu_1}}{\sigma_1\sqrt{k}}\left(\sum_{i=0}^{k}\gamma_i-\frac{k+1}{\mu_1}\right)<t\right\}-\phi(t)\right| = O\left[\left(\frac{ln(k)}{k}\right)^{\frac{1}{4}}\right] \quad \text{(A5)}$$

From this equation, it can be declared that for an arbitrary positive ε_1, there exists a positive integer $N_1(\varepsilon_1)$, such that

$$\left|\mathbf{P}_r\left\{\frac{\sqrt{\mu_1}}{\sigma_1\sqrt{k}}\left(\sum_{i=0}^{k}\gamma_i-\frac{k+1}{\mu_1}\right)<t\right\}-\phi(t)\right|\le\varepsilon_1 \quad \text{(A6)}$$

is valid for every real t, provided that $k\ge N_1(\varepsilon_1)$.

Therefore, when $k\ge N_1(\varepsilon_1)$, the following relations are always valid:

$$-\varepsilon_1\le\mathbf{P}_r\left\{\frac{\sqrt{\mu_1}}{\sigma_1\sqrt{k}}\left(\sum_{i=0}^{k}\gamma_i-\frac{k+1}{\mu_1}\right)<-t\right\}-\phi(-t)\le\varepsilon_1 \quad \text{(A7)}$$

$$-\varepsilon_1\le\mathbf{P}_r\left\{\frac{\sqrt{\mu_1}}{\sigma_1\sqrt{k}}\left(\sum_{i=0}^{k}\gamma_i-\frac{k+1}{\mu_1}\right)<t\right\}-\phi(t)\le\varepsilon_1 \quad \text{(A8)}$$

which further leads to the following inequality, in which t is an arbitrary positive number:

$$\begin{aligned}&\mathbf{P}_r\left\{\left|\frac{\sqrt{\mu_1}}{\sigma_1\sqrt{k}}\left(\sum_{i=0}^{k}\gamma_i-\frac{k+1}{\mu_1}\right)\right|<t\right\}\\&=\mathbf{P}_r\left\{\frac{\sqrt{\mu_1}}{\sigma_1\sqrt{k}}\left(\sum_{i=0}^{k}\gamma_i-\frac{k+1}{\mu_1}\right)<t\right\}\\&\quad-\mathbf{P}_r\left\{\frac{\sqrt{\mu_1}}{\sigma_1\sqrt{k}}\left(\sum_{i=0}^{k}\gamma_i-\frac{k+1}{\mu_1}\right)<-t\right\}\\&\ge[\phi(t)-\varepsilon_1]-[\phi(-t)+\varepsilon_1]\\&=\phi(t)-\phi(-t)-2\varepsilon_1\end{aligned} \quad \text{(A9)}$$

On the other hand, $|(\sqrt{\mu_1}/(\sigma_1\sqrt{k}))(\sum_{i=0}^{k}\gamma_i-(k+1)/\mu_1)|<t$ is equivalent to

$$\left|\sum_{i=0}^{k}\gamma_i-(k+1)\pi_1\right|<\sigma_1 t\sqrt{\pi_1}\sqrt{k} \quad \text{(A10)}$$

which implies that

$$\begin{aligned}\sum_{i=0}^{k}\gamma_i&>(k+1)\pi_1-\sigma_1 t\sqrt{\pi_1}\sqrt{k}\\&=\sqrt{k}\left(\frac{k+1}{\sqrt{k}}\sqrt{\pi_1}-\sigma_1 t\right)\sqrt{\pi_1}\end{aligned} \quad \text{(A11)}$$

Note that $\pi_1>0$ and is independent of k. It is obvious that $((k+1)/\sqrt{k})\sqrt{\pi_1}-\sigma_1 t$ is a monotonically increasing function of k. This means that for an arbitrary positive t, there exists a positive integer $N_2(t)$, such that $((k+1)/\sqrt{k})\sqrt{\pi_1}-\sigma_1 t>0$ is valid for each $k\ge N_2(t)$.

Define $N_2(t)$ and $\xi(t)$, respectively, as

$$N_2(t)=\min\left\{k\,\middle|\,k\text{ is an integer, }\frac{k+1}{\sqrt{k}}\sqrt{\pi_1}-\sigma_1 t>0\right\}$$

$$\xi(t)=\left(\frac{N_2(t)+2}{\sqrt{N_2(t)+1}}\sqrt{\pi_1}-\sigma_1 t\right)\sqrt{\pi_1}$$

Then, it is obvious that for an arbitrary $k\ge N_2(t)+1$, $(((k+1)/\sqrt{k})\sqrt{\pi_1}-\sigma_1 t)\sqrt{\pi_1}\ge\xi(t)>0$, which further leads to

$$\sum_{i=0}^{k}\gamma_i>\sqrt{k}\xi(t) \quad \text{(A12)}$$

In addition, from the definition of the function $\phi(t)$ or the properties of a normal distribution, it can be declared that for an arbitrary $\varepsilon_2>0$, there exists a positive $t(\varepsilon_2)$, such that

$$\phi\left[t(\varepsilon_2)\right]-\phi\left[-t(\varepsilon_2)\right]\ge 1-\varepsilon_2 \quad \text{(A13)}$$

Now, for an arbitrary positive ε, let $\varepsilon_1=\varepsilon/4$ and $\varepsilon_2=\varepsilon/2$. Define $N(\varepsilon)$ as $N(\varepsilon)=\max\{N_1(\varepsilon_1),N_2(t(\varepsilon_2))+1\}$. Then, from (A9) and (A13), we have that when k is larger than $N(\varepsilon)$, it is certain that

$$\mathbf{P}_r\left\{\left|\frac{\sqrt{\mu_1}}{\sigma_1\sqrt{k}}\left(\sum_{i=0}^{k}\gamma_i-\frac{k+1}{\mu_1}\right)\right|<t(\varepsilon_2)\right\}\ge 1-\frac{\varepsilon}{2}-2\times\frac{\varepsilon}{4}=1-\varepsilon \quad \text{(A14)}$$

Based on this relation and (A12), it can be further declared that

$$\begin{aligned}&\mathbf{P}_r\left\{\sum_{i=0}^{k}\gamma_i>\sqrt{k}\xi\left[t\left(\frac{\varepsilon}{2}\right)\right]\right\}\\&\ge\mathbf{P}_r\left\{\left|\frac{\sqrt{\mu_1}}{\sigma_1\sqrt{k}}\left(\sum_{i=0}^{k}\gamma_i-\frac{k+1}{\mu_1}\right)\right|<t(\varepsilon_2)\right\}\\&\ge 1-\varepsilon\end{aligned} \quad \text{(A15)}$$

A combination of this inequality and (A1) makes it clear that if $k\ge N(\varepsilon)$, then, with a probability greater than $1-\varepsilon$

$$\rho_k(X,Y)\le\alpha_{1h}^{\sqrt{k}\xi\left[t\left(\frac{\varepsilon}{2}\right)\right]}\delta(X,Y)=e^{-\sqrt{k}\xi\left[t\left(\frac{\varepsilon}{2}\right)\right]ln\frac{1}{\alpha_{1h}}}\delta(X,Y) \quad \text{(A16)}$$

As $0\le\alpha_{1h}<1$ and $\delta(X,Y)$ is a finite positive number when both X and Y are finite PDMs, it can therefore be declared that $\lim_{k\to\infty}\alpha_{1h}^{\sqrt{k}\xi[t(\varepsilon/2)]}=0$. Recall that $\delta_k(X,Y)$ is nonnegative and ε is an arbitrarily selected positive number, these relations mean that for any finite PDMs X and Y, $\lim_{k\to\infty}\rho_k(X,Y)=0$ in probability. This completes the proof. ◇

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Proof of Theorem 2: Assume that $P^{[j]} = \lim_{k\to\infty} \mathbf{H}_m(M^{[\gamma_0^{[j]}]}M^{[\gamma_{-1}^{[j]}]}\cdots M^{[\gamma_{-k}^{[j]}]}, P^\star)$. Then, for each $* \in \{0,1\}$

$$\begin{aligned}
&\mathbf{H}_m\left(M^{[*]}, P^{[j]}\right)\\
&= \mathbf{H}_m\left(M^{[*]}, \lim_{k\to\infty}\mathbf{H}_m\left(M^{\left[\gamma_0^{[j]}\right]}\right.\right.\\
&\qquad\qquad \left.\left.\times\ M^{\left[\gamma_{-1}^{[j]}\right]}\cdots M^{\left[\gamma_{-k}^{[j]}\right]},\ P^\star\right)\right)\\
&= \mathbf{H}_m\left(M^{[*]}\lim_{k\to\infty} M^{\left[\gamma_0^{[j]}\right]}M^{\left[\gamma_{-1}^{[j]}\right]}\cdots M^{\left[\gamma_{-k}^{[j]}\right]},\ P^\star\right)\\
&= \lim_{k\to\infty}\mathbf{H}_m\left(M^{[*]}M^{\left[\gamma_0^{[j]}\right]}M^{\left[\gamma_{-1}^{[j]}\right]}\cdots M^{\left[\gamma_{-k}^{[j]}\right]},\ P^\star\right)\\
&= \lim_{k\to\infty}\mathbf{H}_m\left(M^{[*]}M^{\left[\gamma_0^{[j]}\right]}M^{\left[\gamma_{-1}^{[j]}\right]}\cdots M^{\left[\gamma_{-k+1}^{[j]}\right]},\right.\\
&\qquad\qquad \left.\mathbf{H}_m\left(M^{\left[\gamma_{-k}^{[j]}\right]}, P^\star\right)\right)\\
&= \lim_{k\to\infty}\mathbf{H}_m\left(M^{[*]}M^{\left[\gamma_0^{[j]}\right]}M^{\left[\gamma_{-1}^{[j]}\right]}\cdots M^{\left[\gamma_{-k+1}^{[j]}\right]}, P^\star\right)\\
&\quad\text{(in prob.)}\\
&= P^{[\bar{j}]}
\end{aligned} \tag{A17}$$

in which $j = n(\gamma_i^{[j]}|_{i=0}^{-\infty}) = \sum_{i=0}^{-\infty}\gamma_i^{[j]}2^i$, $\bar{j} = n(\bar{\gamma}_i^{[j]}|_{i=0}^{-\infty}) = \sum_{i=0}^{-\infty}\bar{\gamma}_i^{[j]}2^i$, and $\bar{\gamma}_i^{[j]} = \gamma_{i+1}^{[j]}$ whenever $i \le -1$ while $\bar{\gamma}_0^{[j]} = *$.

Define $n_{in}(*, \gamma_i^{[j]}|_{i=0}^{-\infty})$ as $n_{in}(*, \gamma_i^{[j]}|_{i=0}^{-\infty}) = \bar{j} - j$. Then

$$\begin{aligned}
n_{in}\left(*, \gamma_i^{[j]}|_{i=0}^{-\infty}\right) &= \sum_{i=0}^{-\infty}\bar{\gamma}_i^{[j]}2^i - \sum_{i=0}^{-\infty}\gamma_i^{[j]}2^i\\
&= \left(* - \gamma_0^{[j]}\right) + \sum_{i=-1}^{-\infty}\left(\gamma_{i+1}^{[j]} - \gamma_i^{[j]}\right)2^i
\end{aligned} \tag{A18}$$

For a given sequence $\gamma_l^{[j]}|_{l=1}^{s}$ with $\gamma_l^{[j]} \in \{0,1\}$, define $\gamma_{i,l}^{[j]}$ as $\gamma_{i,0}^{[j]} = \gamma_i^{[j]}$, $i = 0, -1, \ldots$, and

$$\gamma_{i,l}^{[j]} = \begin{cases}\gamma_{i+1,l-1}^{[j]} & i \le -1\\ \gamma_l^{[j]} & i = 0\end{cases} \qquad l = 1, 2, \cdots, s \tag{A19}$$

Denote $\mathbf{H}_m(M^{[\gamma_l^{[j]}]}M^{[\gamma_{l-1}^{[j]}]}\ldots M^{[\gamma_1^{[j]}]}, P^{[j]})$ by $P^{[j_l]}$, $l = 1, 2, \ldots, s$. Then, a repetitive utilization of (A17) leads to

$$\begin{aligned}
j_s &= j_{s-1} + n_{in}\left(\gamma_s^{[j]}, \gamma_{i,s-1}^{[j]}|_{i=0}^{-\infty}\right)\\
&= j_{s-2} + n_{in}\left(\gamma_{s-1}^{[j]}, \gamma_{i,s-2}^{[j]}|_{i=0}^{-\infty}\right) + n_{in}\left(\gamma_s^{[j]}, \gamma_{i,s-1}^{[j]}|_{i=0}^{-\infty}\right)\\
&= \cdots\\
&= j_0 + \sum_{l=1}^{s} n_{in}\left(\gamma_l^{[j]}, \gamma_{i,l-1}^{[j]}|_{i=0}^{-\infty}\right)
\end{aligned} \tag{A20}$$

in which $j_0 = j$. Therefore, $j_s = j$ if and only if

$$\sum_{l=1}^{s} n_{in}\left(\gamma_l^{[j]}, \gamma_{i,l-1}^{[j]}|_{i=0}^{-\infty}\right) = 0 \tag{A21}$$

On the other hand, from the definition of $n_{in}(*, \gamma_i^{[j]}|_{i=0}^{-\infty})$, straightforward algebraic manipulations show that

$$\begin{aligned}
&\sum_{l=1}^{s} n_{in}\left(\gamma_l^{[j]}, \gamma_{i,l-1}^{[j]}|_{i=0}^{-\infty}\right)\\
&= \sum_{l=1}^{s}\left[\left(\gamma_l^{[j]} - \gamma_{0,l-1}^{[j]}\right) + \sum_{i=-1}^{-\infty}\left(\gamma_{i+1,l-1}^{[j]} - \gamma_{i,l-1}^{[j]}\right)2^i\right]\\
&= \sum_{l=1}^{s}\left(\gamma_l^{[j]} - \gamma_{0,l-s}^{[j]}\right)2^{l-s} + \sum_{i=0}^{-\infty}\left(\gamma_{0,i}^{[j]} - \gamma_{0,i-s}^{[j]}\right)2^{i-s}
\end{aligned} \tag{A22}$$

Therefore, $j_s = j$ if and only if

$$\begin{aligned}
\gamma_s^{[j]} &= \gamma_{0,0}^{[j]} - \sum_{l=1}^{s-1}\left(\gamma_l^{[j]} - \gamma_{0,l-s}^{[j]}\right)2^{l-s}\\
&\quad - \sum_{i=0}^{-\infty}\left(\gamma_{0,i}^{[j]} - \gamma_{0,i-s}^{[j]}\right)2^{i-s}\\
&= \gamma_{0,0}^{[j]} + \sum_{l=1}^{s-1}\left(\gamma_{0,l-s}^{[j]} - \gamma_l^{[j]}\right)2^{l-s}\\
&\quad + \sum_{i=0}^{-\infty}\left(\gamma_{0,i-s}^{[j]} - \gamma_{0,i}^{[j]}\right)2^{i-s}\\
&= \gamma_{0,0}^{[j]} + \sum_{i=-1}^{1-s}\gamma_{0,i}^{[j]}2^i + 2^{-s}\sum_{i=0}^{-\infty}\left(\gamma_{0,i-s}^{[j]} - \gamma_{0,i}^{[j]}\right)2^i\\
&\quad - \sum_{i=1}^{s-1}\gamma_{s-i}^{[j]}2^{-i}
\end{aligned} \tag{A23}$$

Define a set $\mathcal{S}^{[j]}$ as the equation shown at the bottom of the page. Assume that this set is not empty for all the possible j. Then, for any $s \in \mathcal{S}^{[j]}$, there exists a binary sequence $\gamma_i^{[j]}|_{i=1}^{s}$ with $\gamma_i^{[j]} \in \{0,1\}$ such that

$$\mathbf{H}_m\left(M^{\left[\gamma_s^{[j]}\right]}M^{\left[\gamma_{s-1}^{[j]}\right]}\cdots M^{\left[\gamma_1^{[j]}\right]}, P^{[j]}\right) = P^{[j]} \tag{A24}$$

Assume that the Markov chain γ_k is in its stationary state in which both $\mathbf{P}_r(\gamma_k = 1)$ and $\mathbf{P}_r(\gamma_k = 0)$ take a constant value belonging to (0, 1). Denote $\max\{\mathbf{P}_r(\gamma_k = 1), \mathbf{P}_r(\gamma_k = 0)\}$ and $\min\{\mathbf{P}_r(\gamma_k = 1), \mathbf{P}_r(\gamma_k = 0)\}$ respectively by p_{hs} and

$$\mathcal{S}^{[j]} = \left\{k \,\middle|\, k = \min(s),\ \begin{array}{c}\gamma_{0,0}^{[j]} + \sum_{i=-1}^{1-s}\gamma_{0,i}^{[j]}2^i + 2^{-s}\sum_{i=0}^{-\infty}\left(\gamma_{0,i-s}^{[j]} - \gamma_{0,i}^{[j]}\right)2^i - \sum_{i=1}^{s-1}\gamma_{s-i}^{[j]}2^{-i} \in \{0,1\}\\ \gamma_i^{[j]} \in \{0,1\},\ i = 1, 2, \ldots, s-1\end{array}\right\}$$

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p_{ls}. Moreover, for a particular $s \in \mathcal{S}^{[j]}$, denote the corresponding $\gamma_i^{[j]}|_{i=1}^s$ by $\gamma_i^{[j,s]}|_{i=1}^s$. Then

$$\mathbf{P}_r(s)=\prod_{i=1}^{s}\left[\gamma_i^{[j,s]}\mathbf{P}_r\left(\gamma_i^{[j,s]}=1\right)+\left(1-\gamma_i^{[j,s]}\right)\mathbf{P}_r\left(\gamma_i^{[j,s]}=0\right)\right] \tag{A25}$$

Therefore

$$\mathbf{P}_r(s)\geq\prod_{i=1}^{s}\min\left\{\mathbf{P}_r(\gamma_k=1),\mathbf{P}_r(\gamma_k=0)\right\}=p_{ls}^s \tag{A26}$$

$$\mathbf{P}_r(s)\leq\prod_{i=1}^{s}\max\left\{\mathbf{P}_r(\gamma_k=1),\mathbf{P}_r(\gamma_k=0)\right\}=p_{hs}^s \tag{A27}$$

Hence, when an integer s belonging to the set $\mathcal{S}^{[j]}$ takes a finite value, its occurrence probability is certainly greater than 0.

As in Lemma 1, let $\tau_v^{[j]}$ denote the v-th time instant that $j_s = j_0$ and $f_v^{[j]}(P_{k|k})$ the random variable $\sum_{k=\tau_v^{[j]}}^{\tau_{v+1}^{[j]}-1} I_{\mathcal{B}_d}(P_{k|k})$. Then

$$\begin{aligned}\left|f_v^{[j]}(P_{k|k})\right| &= \left|\sum_{k=\tau_v^{[j]}}^{\tau_{v+1}^{[j]}-1} I_{\mathcal{B}_d}(P_{k|k})\right| \leq \left(\tau_{v+1}^{[j]}-1\right)-\left(\tau_v^{[j]}-1\right)\\ &= \tau_{v+1}^{[j]}-\tau_v^{[j]}\in\mathcal{S}^{[j]}\end{aligned} \tag{A28}$$

Hence

$$\begin{aligned}\mathbf{E}\left\{\left|f_v^{[j]}(P_{k|k})\right|^3\right\} &\leq \mathbf{E}\left\{\left(\tau_{v+1}^{[j]}-\tau_v^{[j]}\right)^3\right\} = \sum_{k\in\mathcal{S}^{[j]}} k^3\mathbf{P}_r(k)\\ &\leq \sum_{k=1}^{\infty} k^3 p_{hs}^k\end{aligned} \tag{A29}$$

Note that $k^3 = (k+1)k(k-1)+k$. It can be directly proved that

$$\sum_{k=1}^{\infty} k^3 p_{hs}^k = \frac{(1+p_{hs})^2+2p_{hs}}{(1-p_{hs})^4}p_{hs} \tag{A30}$$

Therefore, when p_{hs} belongs to (0, 1), both $\mathbf{E}\{|f_v^{[j]}(P_{k|k})|^3\}$ and $\mathbf{E}\{(\tau_{v+1}^{[j]}-\tau_v^{[j]})^3\}$ are finite.

Note also that $\mathbf{H}_m(M^{[1]},P^\star)=P^\star$ and $\lim_{k\to\infty}\mathbf{H}_m[M^{[1]k}, \mathbf{H}_m(M^{[0]},P^\star)]=P^\star$. It is obvious that when $j=\sum_{i=0}^{-\infty}2^i$, the set $\mathcal{S}^{[j]}$ has at least two finite integers that has an occurrence probability greater than 0. Therefore, when the random process $P_{k|k}$ is started from $P^\star$, the corresponding $f_\alpha^{[j]} - s(f)(\tau_{\alpha+1}^{[j]} - \tau_\alpha^{[j]})$ has a variance greater than 0.

Denote $\sum_{k\in\mathcal{S}^{[j]}} k\mathbf{P}_r(k)$ by $\mu^{[j]}$. Then, it can be directly proved that

$$F(d)=\sum_{j\in\mathcal{I}}\frac{1}{\mu^{[j]}}I_{\mathcal{B}_d}\left(P^{[j]}\right) \tag{A31}$$

On the other hand, according to Lemma 1, we have that

$$\lim_{n\to\infty}\frac{1}{n+1}\sum_{k=0}^{n}I_{\mathcal{B}_d}(P_{k|k})=\sum_{j\in\mathcal{I}}\frac{1}{\mu^{[j]}}I_{\mathcal{B}_d}\left(P^{[j]}\right) \tag{A32}$$

and the convergence rate is of the order $(ln(n)/n)^{1/4}$. The proof can now be completed for the case in which $\mathcal{S}^{[j]}\neq\emptyset$ for every possible j, through combining the above two equations together.

If there exists a j such that the set $\mathcal{S}^{[j]}$ is empty, the conclusions can still be established through modifying $P^{[j]}$ to $\mathcal{P}^{[j]}(\varepsilon)$ in the above arguments, in which ε is a prescribed positive number. More precisely, according to Theorem 1, for arbitrary j_1 and j_2, there always exists a finite step transformation from an element of $\mathcal{P}^{[j_1]}(\varepsilon)$ to the set $\mathcal{P}^{[j_2]}(\varepsilon)$. Therefore, the corresponding set $\mathcal{S}^{[j]}$ is certainly not empty. The results can then be established through decreasing ε to 0. ◇

Proof of Lemma 5: From $P_{0|0}=P^\star$ and $\mathbf{H}_m(M^{[1]},P^\star)=P^\star$, it is clear that $P_{1|1}$ has only one additional possible value, that is, $\mathbf{H}_m(M^{[0]},P^\star)$. Hence, the number of elements in $\bar{\mathcal{P}}^{[1]}$ is 2 and $\bar{\mathcal{P}}^{[1]}=\{P^\star,\mathbf{H}_m(M^{[0]},P^\star)\}$.

Assume that the conclusions are valid with $n=l$. That is, $\#(\bar{\mathcal{P}}^{[l]})=2^l$ and

$$\begin{aligned}\bar{\mathcal{P}}^{[l]} &= \left\{P^\star,\mathbf{H}_m\left(M^{[0]},P^\star\right)\right\}\\ &\bigcup\left\{P\,\middle|\,\begin{array}{c}P=\mathbf{H}_m\left[M^{[\gamma_k]}M^{[\gamma_{k-1}]}\cdots M^{[\gamma_2]},\mathbf{H}_m\left(M^{[0]},P^\star\right)\right]\\ \gamma_j\in\{0,1\},j\in\{2,3,\cdots,k\},k\in\{2,3,\ldots,l\}\end{array}\right\}\end{aligned} \tag{A33}$$

Then, when $n=l+1$, we have

$$\begin{aligned}&\bar{\mathcal{P}}^{[l+1]}\\ &=\bar{\mathcal{P}}^{[l]}\bigcup\left\{P_{l+1}\,\middle|\,P_{l+1}=\mathbf{H}_m\left(M^{[\gamma_{l+1}]},P\right),\right.\\ &\qquad\left.P\in\bar{\mathcal{P}}^{[l]}\setminus\bar{\mathcal{P}}^{[l-1]},\gamma_{l+1}\in\{0,1\}\right\}\\ &=\bar{\mathcal{P}}^{[l]}\bigcup\left\{P_{l+1}\,|P_{l+1}\right.\\ &\qquad=\mathbf{H}_m\left\{M^{[\gamma_{l+1}]},\mathbf{H}_m\left[M^{[\gamma_l]}M^{[\gamma_{l-1}]}\cdots M^{[\gamma_2]},\right.\right.\\ &\qquad\qquad\left.\left.\mathbf{H}_m\left(M^{[0]},P^\star\right)\right]\right\},\\ &\qquad\left.\gamma_j\in\{0,1\},j\in\{2,3,\ldots,l+1\}\right\}\\ &=\bar{\mathcal{P}}^{[l]}\bigcup\left\{P_{l+1}\,|P_{l+1}\right.\\ &\qquad=\mathbf{H}_m\left[M^{[\gamma_{l+1}]}M^{[\gamma_l]}\cdots M^{[\gamma_2]},\mathbf{H}_m\left(M^{[0]},P^\star\right)\right],\\ &\qquad\left.\gamma_j\in\{0,1\},j\in\{2,3,\ldots,l+1\}\right\}\end{aligned} \tag{A34}$$

From the regularity of the matrices $A^{[0]}$ and $A^{[1]}$, as well as Lemma 3, it can be directly proved that

$$\begin{aligned}&\bar{\mathcal{P}}^{[l]}\bigcap\left\{P_{l+1}|P_{l+1}=\mathbf{H}_m\left[M^{[\gamma_{l+1}]}M^{[\gamma_l]}\cdots M^{[\gamma_2]},\right.\right.\\ &\qquad\left.\left.\mathbf{H}_m\left(M^{[0]},P^\star\right)\right],\gamma_j\in\{0,1\},j\in\{2,3,\ldots,l+1\}\right\}=\emptyset\end{aligned} \tag{A35}$$

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Therefore, $\#(\bar{\mathcal{P}}^{[l+1]}) = \#(\bar{\mathcal{P}}^{[l]}) + 2^{l-1} = 2^l$ and

$$\bar{\mathcal{P}}^{[l+1]} = \left\{P^\star, \mathbf{H}_m\left(M^{[0]}, P^\star\right)\right\} \bigcup \left\{P \middle| \begin{array}{l} P = \mathbf{H}_m\left[M^{[\gamma_k]}M^{[\gamma_{k-1}]}\cdots M^{[\gamma_2]}, \mathbf{H}_m\left(M^{[0]}, P^\star\right)\right] \\ \gamma_j \in \{0,1\}, j \in \{2,3,\ldots,k\}, k \in \{2,3,\ldots,l+1\} \end{array}\right\} \tag{A36}$$

This completes the proof. ◇

Proof of Theorem 3: At first, probabilities are investigated for the occurrence of $P_{k|k} = \bar{P}^{[j]}$ with $j = 1 + 2^{s-1} + \sum_{l=1}^{s-1}\gamma_l^{[j]}2^{l-1}$. From the definition of $\bar{P}^{[j]}$, it is obvious that $P_{k|k} = \bar{P}^{[j]}$ if and only if $k \geq s+1$, $\gamma_l = 1$ when $l \in \{1,2,\ldots,k-s-1\}$, $\gamma_{k-s} = 0$ and $\gamma_{i+k-s} = \gamma_i^{[j]}$ when $i \in \{1,2,\ldots,s\}$. Hence

$$\begin{aligned}
&\mathbf{P}_r\left(P_{k|k} = \bar{P}^{[j]}\right) \\
&= \mathbf{P}_r\Big(\gamma_1 = 1, \ldots, \gamma_{k-s-1} = 1, \gamma_{k-s} = 0, \\
&\qquad \gamma_{k-s+1} = \gamma_1^{[j]}, \ldots, \gamma_k = \gamma_s^{[j]}\Big) \\
&= \prod_{l=1}^{k-s-1}\mathbf{P}_r(\gamma_l = 1) \times \mathbf{P}_r(\gamma_{k-s} = 0)\prod_{i=1}^{s}\mathbf{P}_r\left(\gamma_{i+k-s} = \gamma_i^{[j]}\right) \\
&= \gamma_{st}^{k-s-1}(1-\gamma_{st})p_j
\end{aligned} \tag{A37}$$

in which $p_j = \prod_{i=1}^{s}\mathbf{P}_r(\gamma_{i+k-s} = \gamma_i^{[j]})$.

Therefore, the occurrence of $\bar{P}^{[j]}$ in the samples $P_{0|0}$, $P_{1|1}$, …, $P_{n|n}$ has the following probability $\bar{p}_j$:

$$\begin{aligned}
\bar{p}_j &= \sum_{k=s+1}^{n}\mathbf{P}_r\left(P_{k|k} = \bar{P}^{[j]}\right) \\
&= \sum_{k=s+1}^{n}\gamma_{st}^{k-s-1}(1-\gamma_{st})p_j = \left(1-\gamma_{st}^{n-s}\right)p_j
\end{aligned} \tag{A38}$$

Note that when $\gamma_l^{[j]} \in \{0,1\}, l = 1,2,\ldots,s$, it is certain that $0 \leq \sum_{l=1}^{s-1}\gamma_l^{[j]}2^{l-1} \leq 2^{s-1}-1$. We therefore have that

$$1 + 2^{s-1} \leq j \leq 2^s \tag{A39}$$

which is equivalent to $1 + \log_2(j-1) \geq s \geq \log_2(j)$. As s is a positive integer, it is obvious that

$$s = \lceil \log_2(j) \rceil \tag{A40}$$

Therefore, $\gamma_l^{[j]}$ with $l \in \{1,2,\ldots,\lceil\log_2(j)\rceil\}$ is the binary code of $j - 1 - 2^{\lceil\log_2(j)\rceil - 1}$. It can therefore be declared that for any given j belonging to $\{1,2,\ldots,2^n\}$, both s and $\gamma_l^{[j]}|_{l=1}^{s}$ are uniquely determined through the requirement that $j = 1 + 2^{s-1} + \sum_{l=1}^{s-1}\gamma_l^{[j]}2^{l-1}$.

On the other hand, let $N_0(j)$ denote the number of zeros in the sequence $\gamma_i^{[j]}|_{i=1}^{s}$. Then

$$\begin{aligned}
p_j &= \prod_{i=1}^{s}\mathbf{P}_r\left(\gamma_{i+k-s} = \gamma_i^{[j]}\right) \\
&= \prod_{i=1}^{s}\mathbf{P}_r^{\gamma_i^{[j]}}\left(\gamma_i^{[j]} = 1\right)\mathbf{P}_r^{\left(1-\gamma_i^{[j]}\right)}\left(\gamma_i^{[j]} = 0\right) \\
&= \gamma_{st}^{\sum_{i=1}^{s}\gamma_i^{[j]}}(1-\gamma_{st})^{s-\sum_{i=1}^{s}\gamma_i^{[j]}} \\
&= (1-\gamma_{st})^{N_0(j)}\gamma_{st}^{\lceil\log_2(j)\rceil - N_0(j)}
\end{aligned} \tag{A41}$$

Summarizing (A38), (A40) and (A41), the following formula is obtained for $\bar{p}_j$

$$\bar{p}_j = \left(1-\gamma_{st}^{n-\lceil\log_2(j)\rceil}\right)\gamma_{st}^{\lceil\log_2(j)\rceil - N_0(j)}(1-\gamma_{st})^{N_0(j)} \tag{A42}$$

Note that $N_0(j) = s - \sum_{i=1}^{s}\gamma_i^{[j]}$. Hence, from (A40), the ergodicity of the random process $P_{k|k}$ established in Corollary 1, and the Bernoulli's law of large number [11], it can be claimed that

$$\begin{aligned}
&\lim_{n\to\infty}\frac{1}{n+1}\sum_{k=0}^{n}I_{\mathcal{B}_d}(P_{k|k}) \\
&= \lim_{n\to\infty}\sum_{j\in\mathcal{N}_d}\bar{p}_j \\
&= \lim_{n\to\infty}\sum_{j\in\mathcal{N}_d}\left(1-\gamma_{st}^{n-\lceil log_2(j)\rceil}\right)\gamma_{st}^{\sum_{i=1}^{\lceil log_2(j)\rceil}\gamma_i^{[j]}} \\
&\quad \times (1-\gamma_{st})^{\lceil log_2(j)\rceil - \sum_{i=1}^{\lceil log_2(j)\rceil}\gamma_i^{[j]}}
\end{aligned} \tag{A43}$$

and the convergence rate is exponential. This completes the proof. ◇

Proof of Theorem 4: Recall that for an arbitrary positive integer m, $\mathbf{H}_m(M^{[1]m}, P^\star) = P^\star$. Therefore, in investigating properties of samples of the random process $P_{k|k}$ starting from $P_{0|0} = P^\star$, it can be assumed, without any loss of generality that, the first n elements of any realization of the random process $\gamma_k|_{k=0}^{\infty}$, is a binary sequence satisfying

$$\begin{aligned}
&\gamma_{t_j} = \gamma_{t_j+1} = \cdots = \gamma_{t_j+m_j} = 0,\ m_j \geq 0, \\
&j = 1,2,\ldots,p; \gamma_{t_j+m_j+1} = \gamma_{t_j+m_j+2} = \cdots = \gamma_{t_{j+1}-1} = 1, \\
&1 = t_1 < t_2 < \cdots < t_p \leq n
\end{aligned}$$

Denote by $\hat{\mathcal{P}}^{[n]}$ the set consisting of the first n $P_{k|k}$s generated from this particular realization of the data loss process.

Concerning the Riccati recursion of (4), the following inequality is obtained from (19) for every positive integer m and every positive definite X

$$\begin{aligned}
&\delta\left[\mathbf{H}_m\left(M^{[0]m}, X\right), P^\star\right] \\
&\quad = \delta\left\{\mathbf{H}_m\left[M^{[0]}, \mathbf{H}_m\left(M^{[0](m-1)}, X\right)\right], P^\star\right\} \\
&\quad \leq a\delta\left[\mathbf{H}_m\left(M^{[0](m-1)}, X\right), P^\star\right] + b \\
&\quad = \cdots \\
&\quad \leq a^m\delta(X, P^\star) + a^{m-1}b + a^{m-2}b + \cdots + b \\
&\quad = a^m\delta(X, P^\star) + \frac{1-a^m}{1-a}b
\end{aligned} \tag{A44}$$

Moreover

$$\begin{aligned}
&\delta\left[\mathbf{H}_m\left(M^{[1]m}, X\right), P^\star\right] \\
&\quad = \delta\left[\mathbf{H}_m\left(M^{[1]m}, X\right), \mathbf{H}_m\left(M^{[1]m}, P^\star\right)\right] \\
&\quad \leq \alpha_{1h}^m\delta(X, P^\star)
\end{aligned} \tag{A45}$$

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On the basis of these two inequalities, the following inequality is established for each $j \geq 1$ and every k belonging to $\{t_j + m_j + 1, t_j + m_j + 2, \cdots, t_{j+1} - 1\}$:

$$\begin{aligned}
&\delta(P_{k|k}, P^\star) \\
&= \delta\left[\mathbf{H}_m\left(\prod_{i=k}^{1} M^{[\gamma_i]}, P^\star\right), P^\star\right] \\
&= \delta\Big[\mathbf{H}_m\Big(M^{[1](k-t_j-m_j)} M^{[0](m_j+1)} M^{[1](t_j-t_{j-1}-m_{j-1})} \cdots \\
&\qquad M^{[0](m_1+1)}, P^\star\Big), P^\star\Big] \\
&\leq \alpha_{1h}^{k-t_j-m_j} \delta\left[\mathbf{H}_m\left(M^{[0](m_j+1)} M^{[1](t_j-t_{j-1}-m_{j-1})} \cdots \right.\right. \\
&\qquad \left.\left. M^{[0](m_1+1)}, P^\star\right), P^\star\right] \leq \alpha_{1h}^{k-t_j-m_j} \\
&\times\Big\{a^{m_j+1} \delta\Big[\mathbf{H}_m\Big(M^{[1](t_j-t_{j-1}-m_{j-1})} M^{[0](m_{j-1}+1)} \cdots \\
&\qquad M^{[0](m_1+1)}, P^\star\Big), P^\star\Big] + \frac{1-a^{m_j+1}}{1-a} b\Big\} \\
&\leq \cdots \leq \alpha_{1h}^{k-t_j-m_j} \frac{1-a^{m_j+1}}{1-a} b \\
&+ \alpha_{1h}^{k-t_{j-1}-(m_j+m_{j-1})} \frac{1-a^{m_{j-1}+1}}{1-a} b + \cdots \\
&+ \alpha_{1h}^{k-t_1-\sum_{i=1}^{j} m_i} \frac{1-a^{m_1+1}}{1-a} b
\end{aligned} \tag{A46}$$

Recall that $0 \leq \alpha_{1h} < 1$. We therefore have that if $0 \leq a < 1$, then

$$\begin{aligned}
&\delta(P_{k|k}, P^\star) \\
&\leq \alpha_{1h}^{k-t_j-m_j} \frac{b}{1-a} + \alpha_{1h}^{k-t_{j-1}-(m_j+m_{j-1})} \frac{b}{1-a} + \cdots \\
&\quad + \alpha_{1h}^{k-t_1-\sum_{i=1}^{j} m_i} \frac{b}{1-a} \\
&= \alpha_{1h}^{k-t_j-m_j} \frac{b}{1-a}\left[1 + \alpha_{1h}^{t_j-t_{j-1}-m_{j-1}} + \cdots \right. \\
&\qquad \left. + \alpha_{1h}^{t_j-t_1-\sum_{i=1}^{j} m_i}\right] \\
&= \alpha_{1h}^{k-t_j-m_j} \frac{jb}{1-a}
\end{aligned} \tag{A47}$$

Moreover, if $a = 1$, then

$$\begin{aligned}
\delta(P_{k|k}, P^\star) &\leq \alpha_{1h}^{k-t_j-m_j}(m_j+1)b + \alpha_{1h}^{k-t_{j-1}-(m_j+m_{j-1})} \\
&\quad \times (m_{j-1}+1)b + \cdots + \alpha_{1h}^{k-t_1-\sum_{i=1}^{j} m_i}(m_1+1)b \\
&\leq \alpha_{1h}^{k-t_j-m_j} b\Big\{(m_j+1) + \alpha_{1h}^{t_j-t_{j-1}-m_{j-1}}(m_{j-1}+1) \\
&\quad + \cdots + \alpha_{1h}^{t_j-t_1-\sum_{i=1}^{j-1} m_i}(m_1+1)\Big\} \\
&\leq \alpha_{1h}^{k-t_j-m_j} jb \max_{1\leq i\leq j}(m_i+1)
\end{aligned} \tag{A48}$$

Furthermore, if $a > 1$, then

$$\begin{aligned}
\delta(P_{k|k}, P^\star) &\leq \alpha_{1h}^{k-t_j-m_j} \frac{a^{m_j+1}}{a-1} b \\
&\quad + \alpha_{1h}^{k-t_{j-1}-(m_j+m_{j-1})} \frac{a^{m_{j-1}+1}}{a-1} b \\
&\quad + \cdots + \alpha_{1h}^{k-t_1-\sum_{i=1}^{j} m_i} \frac{a^{m_1+1}}{a-1} b \\
&\leq \alpha_{1h}^{k-t_j-m_j} \frac{ab}{a-1}\Big\{a^{m_j} + \alpha_{1h}^{t_j-t_{j-1}-m_{j-1}} a^{m_{j-1}} \\
&\qquad + \cdots + \alpha_{1h}^{t_j-t_1-\sum_{i=1}^{j-1} m_i} a^{m_1}\Big\} \\
&\leq \alpha_{1h}^{k-t_j-m_j} \frac{jab}{a-1} \max_{1\leq i\leq j} a^{m_i}
\end{aligned} \tag{A49}$$

On the other hand, for any $k \in \{t_j, t_j+1, \ldots, t_j+m_j\}$ with $j = 1, 2, \ldots, p$, from Lemma 3 and (5), as well as $0 \leq \alpha_{0h} \leq 1$, we have that

$$\begin{aligned}
&\delta\left[P_{k|k}, \mathbf{H}_m\left(M^{[0](k-t_j+1)}, P^\star\right)\right] \\
&\leq \delta\left[\mathbf{H}_m\left(\prod_{i=k}^{t_j} M^{[\gamma_i]}, P_{t_j-1|t_j-1}\right), \mathbf{H}_m\left(M^{[0](k-t_j+1)}, P^\star\right)\right] \\
&= \delta\Big[\mathbf{H}_m\Big(M^{[0](k-t_j+1)}, P_{t_j-1|t_j-1}\Big), \mathbf{H}_m\Big(M^{[0](k-t_j+1)}, P^\star\Big)\Big] \\
&\leq \alpha_{0h}^{k-t_j+1} \delta\left(P_{t_j-1|t_j-1}, P^\star\right) \\
&\leq \delta\left(P_{t_j-1|t_j-1}, P^\star\right)
\end{aligned} \tag{A50}$$

Note that $\mathbf{H}_m(I, P^\star) = P^\star$. It can therefore be declared from (A46)–(A50) that

$$\lim_{\alpha_{1h}\to 0} \sup_{P\in\hat{\mathcal{P}}^{[n]}} \inf_{Q\in\tilde{\mathcal{P}}^{[n]}} \delta(P, Q) = 0 \tag{A51}$$

The proof can now be completed by noting that the above conclusion is valid for all the possible $(t_j, m_j)|_{j=1}^{p}$ and p. ◇

Proof of Theorem 5: When the assumption is satisfied, assume that $\mathbf{P}_r\{P_{\infty|\infty} = \mathbf{H}_m(M^{[0]i}, P^\star)\} = a_i$, $i = 0, 1, \ldots$. Then, from the definition of probabilities, we have that

$$\sum_{i=0}^{\infty} a_i = 1 \tag{A52}$$

On the other hand, note that $\mathbf{H}_m[M^{[0]}, \mathbf{H}_m(M^{[0]i}, P^\star)] = \mathbf{H}_m(M^{[0](i+1)}, P^\star)$. Moreover, when the Markov chain achieves its stationary state, $\mathbf{P}_r(\gamma_k = 1) = \gamma_{st}$. It can therefore be declared that when the random process $P_{k|k}$ reaches its stationary state

$$\begin{aligned}
&\mathbf{P}_r\left\{P_{n+1|n+1} = \mathbf{H}_m\left(M^{[0](i+1)}, P^\star\right)\right\} \\
&\qquad = (1-\gamma_{st})\mathbf{P}_r\left\{P_{n|n} = \mathbf{H}_m\left(M^{[0]i}, P^\star\right)\right\}
\end{aligned} \tag{A53}$$

Moreover, to guarantee the stationarity of the random process, it is necessary that

$$\begin{aligned}
&\lim_{n\to\infty} \mathbf{P}_r\left\{P_{n+1|n+1} = \mathbf{H}_m\left(M^{[0]i}, P^\star\right)\right\} \\
&\qquad = \lim_{n\to\infty} \mathbf{P}_r\left\{P_{n|n} = \mathbf{H}_m\left(M^{[0]i}, P^\star\right)\right\}
\end{aligned} \tag{A54}$$

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Therefore

$$a_{i+1} = (1-\gamma_{st})a_{i-1},\ i = 1, 2, \cdots \tag{A55}$$

Substitute this relation into (A52), the following equation is obtained:

$$a_0 + (1-\gamma_{st})a_0 + (1-\gamma_{st})^2 a_0 + \cdots = 1 \tag{A56}$$

Hence

$$a_0 = \frac{1}{\sum_{i=0}^{\infty}(1-\gamma_{st})^i} = \gamma \tag{A57}$$

which further leads to

$$\mathbf{P}_r\left\{P_{\infty|\infty} = \mathbf{H}_m\left(M^{[0]i}, P^\star\right)\right\} = a_0(1-\gamma_{st})^i = \gamma_{st}(1-\gamma_{st})^i \tag{A58}$$

This completes the proof. ◇

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